

Solve It! **Mysteries** for enquiring minds

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Karl Jacklin

A G.A.T.E.WAYS Challenge



Quality ready-to-use
resources



Solve It!

Mysteries

for enquiring minds



Karl Jacklin
A G.A.T.E.WAYS Challenge



Karl Jacklin has been a teacher, school principal, Manager of Personnel and Administration in the North West Metropolitan Region of the Victorian Education Department, a teachers' college lecturer, tutor in statistics and measurement at Deakin University, and manager of his own software development and computer consultancy company. He has also presented many highly successful programs with G.A.T.E.WAYS (Gifted and Talented Education, Extension and Enrichment), an independent, non-funded organisation which aims to provide opportunities for children of 'like-minds' to participate in programs that will challenge them intellectually, and help develop their individual talents.

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G.A.T.E.WAYS directors, Win Smith and Jill Lawrence, wish to express their deep appreciation to Karl for his dedication in writing this exciting and challenging publication.

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Introduction



Solve it! Mysteries for enquiring minds is a collection of 10 hypothetical scenarios across a range of challenging topics. It is designed to encourage students to use problem-solving techniques and logical reasoning, and to consolidate and test their comprehension and writing skills.

Five scenarios are contained in this book, and five more are in Book 2. Each one includes teacher notes which provide a background briefing, related activities and sample answers, and photocopiable information sheets which students will need to solve the mystery.

Each unit of work is suitable for individual or group work within the regular classroom, or team competitions and extension programs.



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How to use the mysteries

There are five mysteries in Book 1 and a further five in Book 2.

Each of the 10 scenarios:

- 1** outlines an event requiring a logical explanation, based on the information presented in the student resource material. The final explanation may be definitive, such as *The murderer was ...*, or it may be the most appropriate of several plausible alternatives.

Students should be encouraged to ask: *What does this evidence tell me? How does it logically lead to a solution? What other evidence might be useful in reaching a conclusion?*

Some students will arrive at the correct solutions to the mysteries, but not adequately explain how they reached their conclusions. Their answers will be based more on intuition than on a logical and connected line of reasoning. Intuition and guesswork can be an important part of problem-solving and should not be discouraged – they can be the basis of a more formal line of investigation. However, students do need to be encouraged to provide evidence of a reasoned solution process

- 2** provides a basis for discussion about the mechanics of research, reading comprehension, clear expression and analytical and deductive reasoning. The various activities each scenario contains are designed to enable students to put into practice such skills as:
 - sorting out the main issues
 - distinguishing between what is relevant and what are merely ‘red herrings’
 - identifying plausible alternatives against which to test one’s evidence

- 3** is self-contained, in that no additional information is required to reach a satisfactory conclusion

- 4** is open-ended and can be used by you and your students as a prompt for further research, exploration and discussion

- 5** can be used to initiate a range of associated class activities. Students might debate a broader version of the topic for instance, basing their arguments on an analysis of material found in the media, books or on the Internet. Or they may respond to the topic with pieces of creative writing or role plays

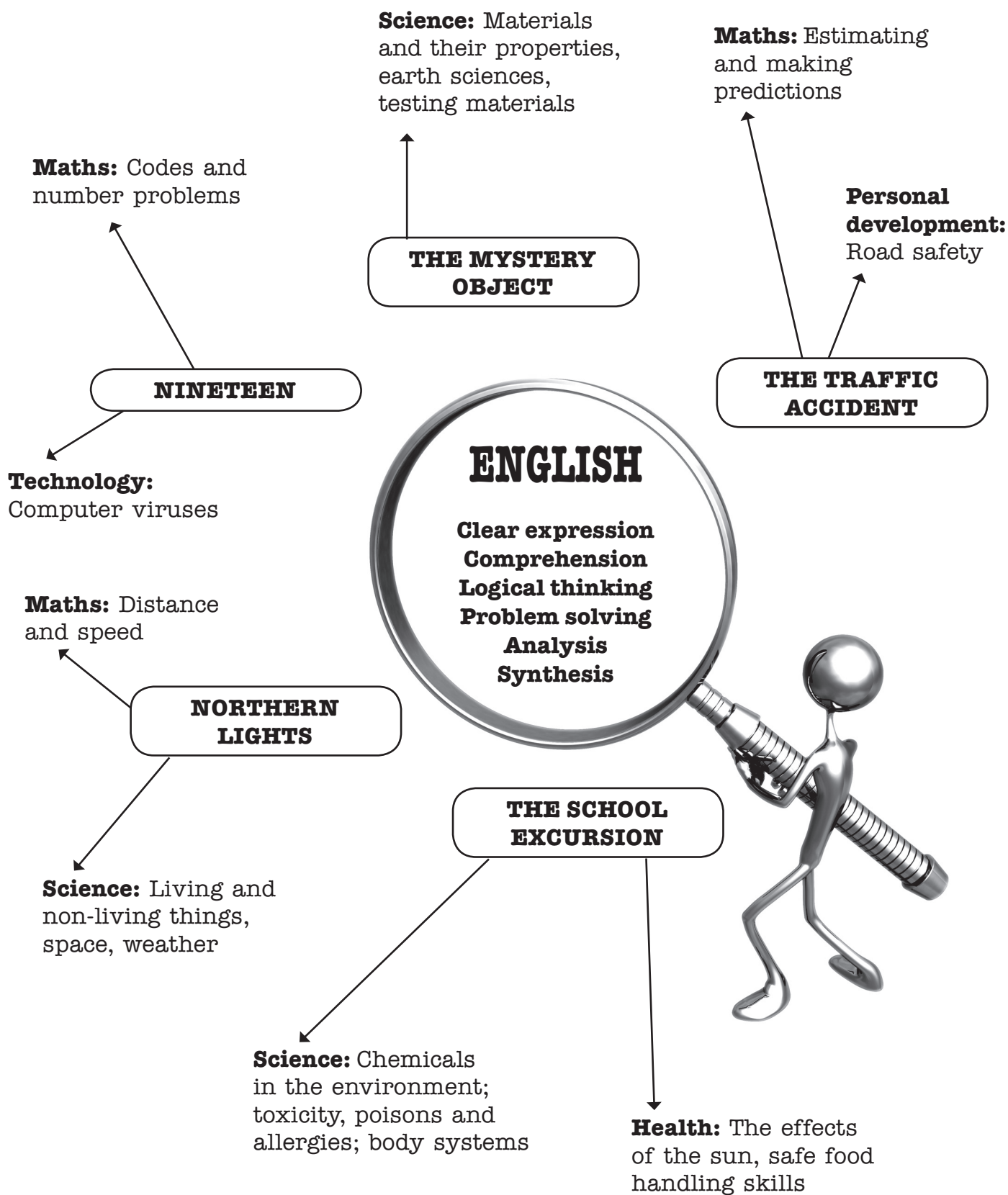
- 6** provides an opportunity to study and employ scientific, logical, and evidence-based methods of evaluating controversial issues such as the presence of alien visitors on Earth or paranormal activity

- 7** includes tasks that enable teachers to integrate knowledge and skills from several curriculum areas into the scenario.



An integrated curriculum

CURRICULUM LINKS





Suggested approach and timing

How you develop a topic is entirely up to you. Each of these 10 mysteries deals with an issue of interest that can be enjoyable in its own right. The mysteries can, however, also be the source of a lot of other intellectually stimulating and challenging activities involving research, debate, discussion, role play, mathematical analysis, scientific experimentation and creative writing.

Teachers who have used this material report that without further extension activities, their classes have completed the tasks within an hour to an hour and a half.

Prior to the activity, students could be encouraged to find out something about the underlying topic, such as an item from the newspaper or some information from the school or local library or the Internet. This information could then be collated during a class discussion, noting the things the students generally agree on about the topic, and highlighting the issues that remain unanswered or are in dispute.

It will become evident during these discussions that the status of the sources of the information provided by the students is of particular importance. Teachers can take this opportunity to discuss issues to do with the validity and accuracy of material they research, as well as the purpose and intent of the author.

TEACHER'S NOTES PAGES

Each scenario includes suggestions as to how best approach the mystery, as well as more detailed descriptions about the skills and knowledge that are explored within it.

Teachers are also guided in the types of questions that they can use to assist students to start the investigation process. For instance, once the topic has been clearly outlined, students should define:

- what they know about the mystery from what they have read so far
- what they are unsure about
- what they would like to or need to know more about.

SAMPLE ANSWERS

A set of sample answers to the questions posed on the student report is provided in the teacher's notes for each mystery. The sample answers are not intended to be definitive. They provide a suggested level of detail that could be expected in a final report by students and will assist teachers with setting any marking criteria or assessment.

STUDENT INFORMATION SHEETS

Complete sets of photocopiable student pages need to be prepared before the lesson. Make sure that the pages are readily available to each student if they are working in groups, as sometimes sharing the information pages inhibits students from knowing all the evidence available or understanding the complete scenario. Students should be encouraged to organise the data they obtain from these sheets, using devices such as tables, Venn diagrams or concept maps to help them logically consider and analyse all the available evidence.

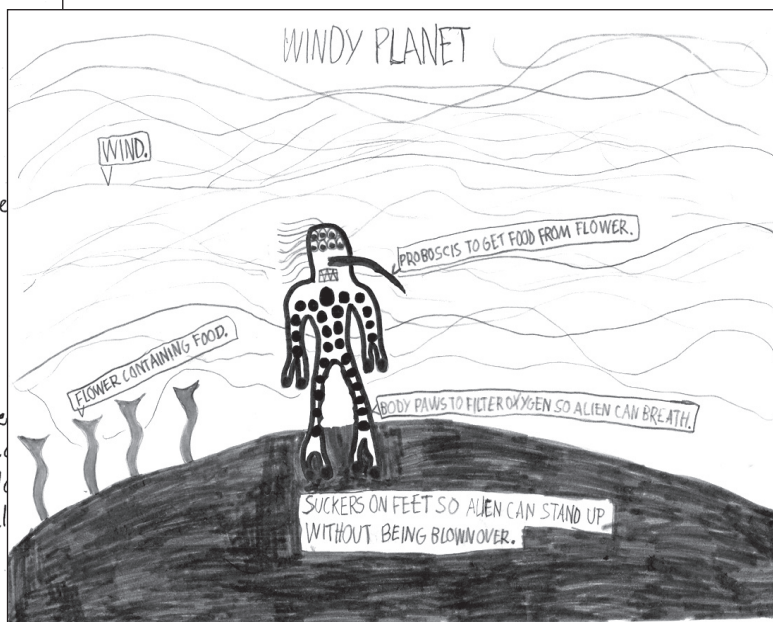
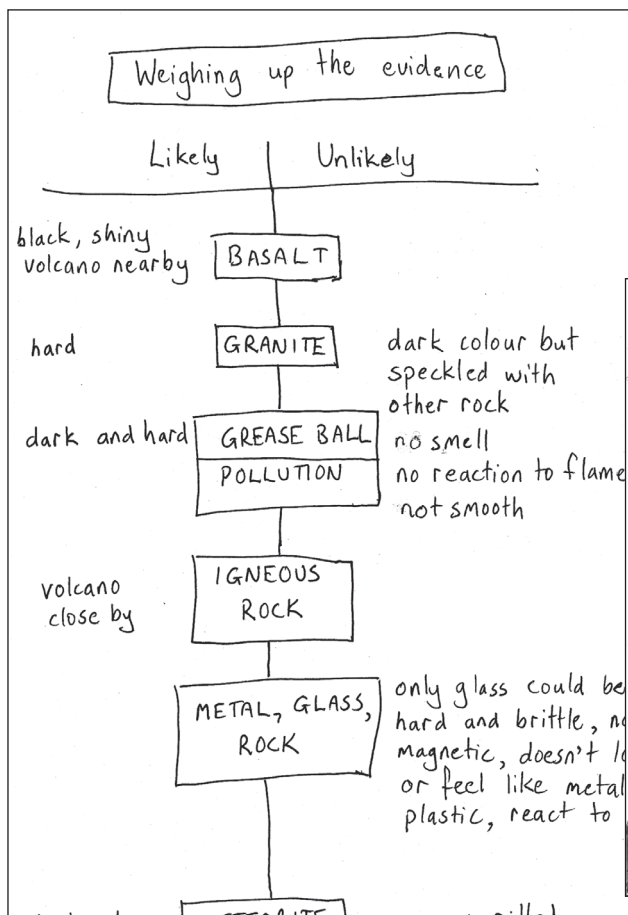
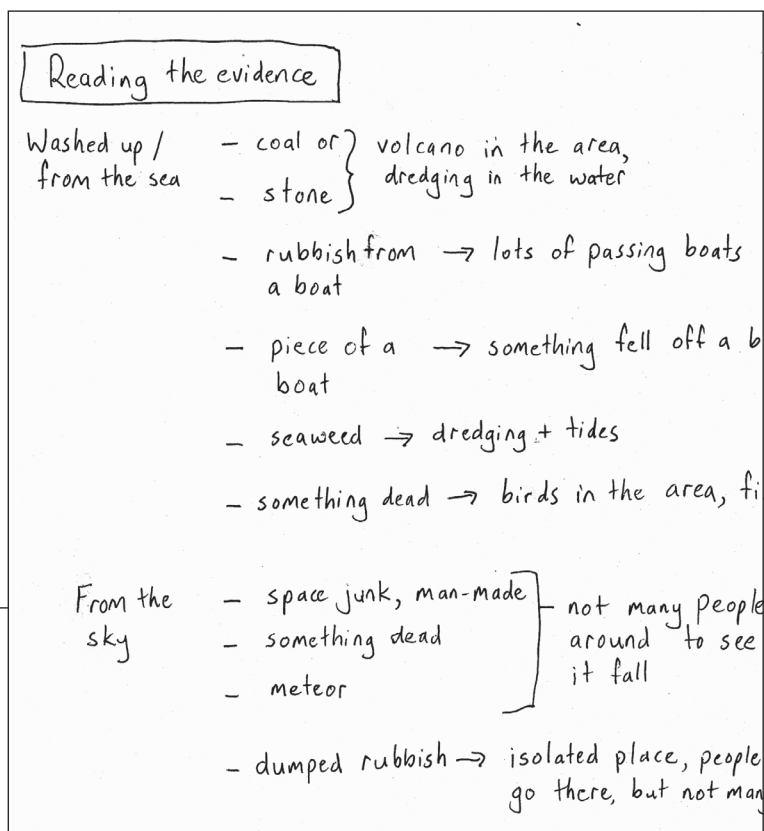
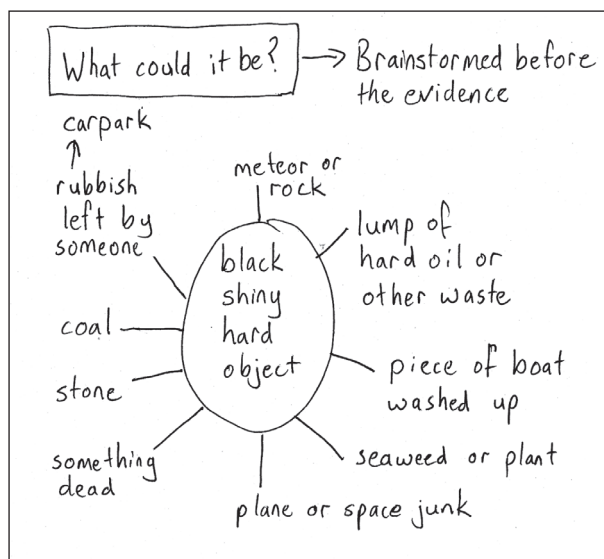


Suggested approach and timing

USING DIAGRAMMATIC TOOLS TO ORGANISE AND ANALYSE THE SCENARIOS

Encourage students to use a range of diagrammatic tools to help them analyse the evidence that is presented to them in each scenario. Tools such as using the DOVE guidelines, Y charts, sunshine wheels, concept and mind maps, Venn diagrams, flow diagrams and the K-W-L-H organiser will help students to develop individual explanations for events and provide a reference for them to write about their choices.

The samples of student work below show three stages in a student's response to *The Mystery Object* scenario in this book. At the bottom of the page there is an illustrated student response to the 'Try this' activity in *Northern Lights*.



1 The mystery object

TEACHER'S NOTES

This activity is a test of students' ability to read with understanding, and to give concise, coherent explanations for a particular choice, rather than to unravel a complex mystery.

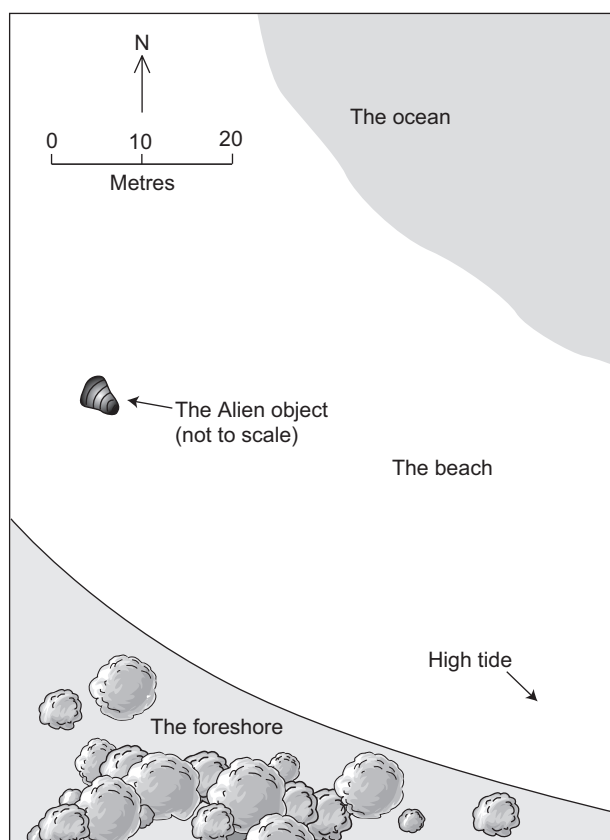
A black, shiny object small enough to fit in the palm of your hand is found on a deserted section of beach. What is it, and from where did it originate?

The question is, what is it?

Well it could be ...

- of a synthetic or natural origin
- a pollutant
- an industrial waste product
- a dead creature
- seaweed ...

or any of a wide range of other possibilities.



The objective at the start is to get the students thinking broadly about alternatives (without referring to any further information) by using their imaginations and a reasonable amount of logical and deductive reasoning. This ensures that the method by which they reach a final conclusion will not be too narrow. Some students are likely to select an answer from a small range of alternatives that do not include the correct answer. Students need to realise that

it is pointless including an object if they cannot think of any possible way that it might have got there.

The more likely alternatives are suggested, at least in part, by the location. There is a sewage treatment plant and a plastics manufacturing plant not far away, oil tankers sailing in and out, and a long extinct volcano nearby. However, someone is likely to suggest that the object might have been dropped there by a tourist or a local resident. This is a reasonable suggestion that warrants a few follow-up questions. For example, are there many visitors to the beach? What sort of rubbish is normally left lying around recreational beaches? Why would someone bring such an object to the beach and then leave it? Does the object look as though it has been worn down by the elements over a long period?

So where did it come from?

It is possible that it was:

- washed up on the beach
- washed down onto the beach
- on the beach and under the sand for thousands of years, and finally exposed by the effects of wind and water.

Students should narrow down this list to just one object, based on the information provided. If still in doubt, they should seek additional information from other resource material.

RELATED ACTIVITIES

Students may find it useful to complete some or all of these activities before the scenario, especially the first one.

Scientific concepts

The meaning of terms such as 'hard', 'brittle' and 'dense' could be explored further. For example, can something be hard without being brittle? Alternatively, can a thing be brittle and not hard? Are hard and/or brittle objects necessarily dense? Lead is one of the denser metals, yet it is quite soft. What causes something to become brittle? How might you test the brittleness of an object? It is said that diamond is one of the hardest naturally occurring substances in the world. How then do you explain the fact that you can smash a diamond to powder with a hammer, and yet barely leave a mark on a piece of steel?

1 The mystery object

TEACHER'S NOTES

Magnetism

There are a whole range of activities relating to magnetism and electrical conductivity that can be explored. Multimeters can be used to test conductivity. Compasses are fairly sensitive to magnetic fields. Students can collect different types of rocks and test them. A meteorite is composed in part of iron. Is it likely to be magnetic? What makes something magnetic? Why do all metals conduct electricity, but only iron can be magnetised? If that's true, how do fridge magnets work?

Volcanoes

There is also much research and activity work that can be undertaken about volcanoes. This topic provides opportunities in the sciences of geology, chemistry and physics.

Time

On an excursion to an old growth forest in Tasmania several years ago, the author was told by the guide as he pointed towards a particular Huon pine that it was around 1000 years old. It had begun its life in a world vastly different from the one in which it now lived. The lump of obsidian found on the beach in this activity had been violently hurled from a volcano almost 10 000 years ago. Imagine if that piece of volcanic rock could tell its story. What interesting events might it have witnessed during the intervening time? Would the piece of obsidian be old enough to have witnessed the existence of the mega fauna that are believed to have roamed parts of Australia? When was the last ice age? Have there been any significant geological changes during this time?

SAMPLE REPORT

Note: The marking scheme is a suggestion only. You may not want to score the work at all.

List some possible materials that the mystery object could be (15 points)

- A seashell (normally easily recognised)
- A shark egg (not nearly as well recognised)
- A tar ball (similar objects are often found near road construction sites)
- A lump of seaweed (normally easily recognised)

- A grease ball (not common and probably not easily recognised)
- A piece of rock (recognisable, but not particularly alien in appearance)
- A meteorite (may not be distinguishable from many rocks)
- Space debris (may not be recognisable as such)
- A lump of plastic

What actually IS the mystery object? (5 points)

A piece of obsidian – a shiny, jet-black volcanic rock.

What are the reasons for your selection? (25 points)

- The object has the outward physical characteristics of obsidian – it is black and shiny and smooth like glass. This excludes the seaweed, shark egg and most seashells.
- It is hard, brittle and dense, which eliminates the possibility of it being a tar ball or a grease ball, as does the fact that it is non-flammable. This also suggests that it is not plastic, which you would expect to respond to heat in some way (melt and/or char).
- Its lack of odour probably eliminates the possibility of it being a grease ball, which you would expect to have some bacterial contamination and therefore be 'smelly'.
- It does not conduct electricity, which indicates that it has no metallic content.
- It is non-magnetic and does not respond to a strong magnet, which suggests that it contains no iron.
- It has the properties of a volcanic rock. This is supported by the fact that it has been found in reasonably close proximity to an extinct volcano.

Do you think that the object has any particular value, and if so, what should be done with it? (5 points)

A piece of obsidian might be of interest to geologists, vulcanologists, and some collectors, but it is fairly common and would only be of significance if it had turned up in an unexpected location. The extinct volcano nearby suggests that the area probably abounds in volcanic rock.

The mystery object



YOUR TASK - You are a scientist with the Government Investigation Unit and you have been instructed to investigate the discovery of a mysterious object found on a local beach. You must identify the object and determine how it came to be there. Then decide whether it has any scientific significance, and recommend what should be done with it.

This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

- 1 Examine the illustration of the object and read the description of the area in which it was found. Then study the maps and write on your report sheet a list of things that you think the mystery object could plausibly be.
- 2 Read the report about the object by the person who found it, Jackson Brown. Then you may want to revise your list of possibilities.
- 3 In order to work out what the object actually is, you will need to do some research. Jackson's test results and the notes on the properties of some possible objects provide the information you need for this. Read these and then deduce which of the listed objects it is most likely to be. Write your answer on the report sheet.
- 4 Provide the reasons for your selection on the report sheet. This is an important part of your task, for even if you have made an incorrect choice in step 3 above, the quality of your reasoning is still important.
- 5 Finally, write whether you think the object has any scientific significance and recommend what should be done with it.

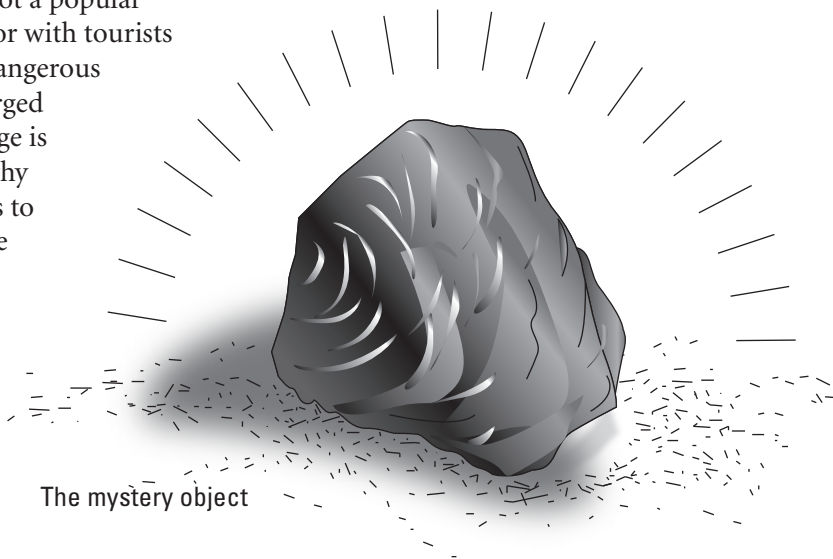
A DESCRIPTION OF THE AREA IN WHICH THE OBJECT WAS FOUND

The beach on which the object was found is not a popular recreational area among the local residents, nor with tourists in general. It tends to be isolated because of dangerous offshore currents and rips caused by a submerged basalt reef, and the fact that the town's sewerage is vented into the sea in this area. This may be why the local council allowed two major industries to be established so close to an otherwise pristine part of the coastline.

A deep channel has also been dredged, providing docking access to oil tankers and other cargo vessels. A large proportion of the locals are employed by these two multi-national companies. The council has strict environmental by-laws, but occasionally there is evidence of pollution.

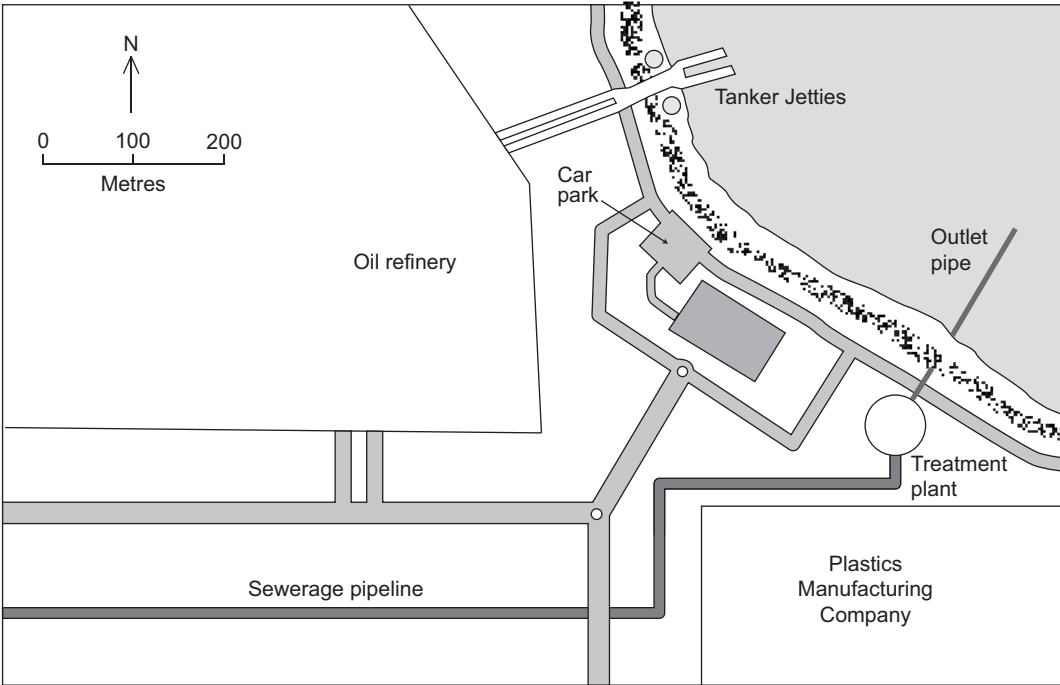
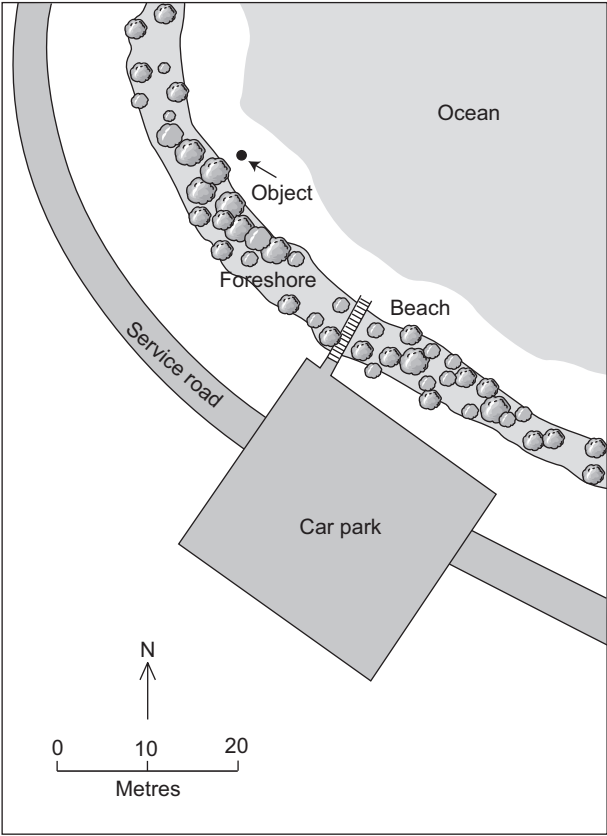
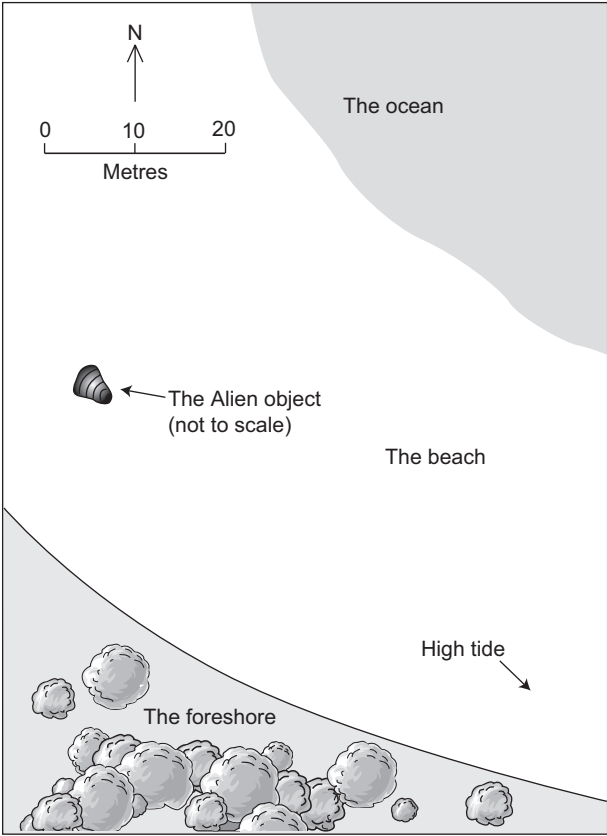
Because of its isolation, the beach is sometimes illegally used by beach buggy enthusiasts, and irresponsible locals have been known to use it as a convenient rubbish dump. However, this isolation has also had a positive impact. Many species of native bird life inhabit the area, and it is becoming well known to bird watchers and other conservationists.

The land is not especially suitable for farming, as it is very rocky. Lava from a long extinct volcano around thirty kilometres from the coast has left a legacy of igneous rock right down to the sea.



The mystery object

The mystery object



The mystery object



REPORT BY JACKSON BROWN, AGED 13 YEARS

I was walking along the beach looking for seashells to add to my collection when I saw this strange lump of something, lying on the beach. At first I thought it was a piece of seaweed, but as I got closer I realised it was quite solid. It was hard, black and shiny, like nothing I'd ever seen before. I picked it up, but then quickly dropped it again, in case it was dangerous. What if it was radioactive or the toxic by-product from some petro-chemical production process?

Not far from where I live there's an oil refinery and a large plastics manufacturing plant, and it occurred to me that the object might have come from one of those two places, or even from a passing oil tanker sluicing out its storage tanks. For all I knew, crude oil might solidify in sea water and turn out like this.

Then again, it might have been space debris of some sort, or a meteorite, or a piece of a rocket or satellite that plunged to Earth through the atmosphere, fused into a molten mass by friction. It might even have fallen from a passing passenger jet. If it had come from a satellite, it might be radioactive – I thought I'd read that they were usually atomic powered.

Then there's the possibility that it was the egg of some sea creature, perhaps a shark – though I guess this is less likely. Or maybe it was just a rock, but it didn't look like a normal one.

I realised that I would have to do some research, but first I had to get it home. Then I'd borrow some materials and equipment from the school's science lab and test some of the object's properties. I didn't want to pick it up again in my hands and carry it, so I pushed it with a piece of wood into my empty backpack, and dragged it home behind me. For safety, I stored it in a metal box in the toolshed at the end of the yard where I could examine it closely. The results of my tests follow.



RESULTS OF JACKSON BROWN'S TESTS ON THE MYSTERY OBJECT

Test	Observations and results
Appearance	Black and shiny
Texture	Smooth like glass
Hardness	Very hard and brittle
Density	Very dense – does not float in water
Combustibility	Does not react to heat or flame
Radioactivity	Does not register on the Geiger counter
Magnetism	Does not show any evidence of magnetism
Electrical conductivity	Does not conduct electricity
Odour	Has no noticeable odour
Other comments	Breaking a piece off the object demonstrates that the colour extends throughout the material

The mystery object



PROPERTIES OF POSSIBLE OBJECTS

Background radiation: Low-level radiation occurring naturally as a result of radioactivity present in the air, soil, buildings and other structures.

Granite: A hard, dark grey, coarse-grained, igneous rock made up of feldspar, mica and quartz.

Grease balls: The discharge into the sea of untreated sewage containing effluent, household detergents, fatty products such as kitchen grease and oils, as well as industrial hydrocarbons (motor oils, grease and rubber) can result in the formation of grease balls. These are combinations of the type of substances mentioned above, and result from tidal forces acting on the mixture like a slow blender. Grease balls can be dangerous, as faecal bacteria may be protected from the seawater by the insoluble outer layer of solidified fats and greases. Over time such objects darken and harden. They are not especially combustible, but they will burn if subjected to sufficient heat.

Igneous rocks: Rocks that are formed under conditions of intense heat or produced by the solidification of volcanic magma on or below Earth's surface.

Metal, glass or plastic: Although an unlikely explanation, the object might have been part of some industrial rubbish illegally dumped on the foreshore that washed down onto the beach during a king tide. A piece of metal or glass previously subjected to sufficient heat might take on the appearance of the object, as might a lump of shiny, black plastic. They might be quite hard – depending on the manufacturing processes involved – but there would be differences in their density and brittleness.

Meteorite: A meteorite is a piece of rock that has reached Earth from outer space. Meteors generally burn up on entering Earth's atmosphere (observed as shooting stars) but occasionally larger ones make it to Earth's surface. They are usually dark grey in colour, very hard, and relatively smooth, although sometimes pitted. There are several different types of meteorite, although one form – stony-iron meteorites – contain silicon-based stone and iron-nickel metal.

Obsidian: A shiny, jet-black, volcanic rock, chemically similar to granite and formed by the rapid cooling of molten lava that produces glass rather than crystals. It is fairly brittle.

Radioactive materials: These contain nucleotides – atoms that are decaying and releasing radiation in the form of particles and rays. Radiation is not by definition dangerous to human health. It depends on the type of radiation, its intensity and exposure time.

Seaweed: Often seaweed washed up on beaches dries and hardens into solid, dark clumps. Even so, it would be difficult to mistake a piece of seaweed for (say) a meteorite. Seaweed is combustible and usually has a characteristic odour and texture.

Shark eggs: These are occasionally washed up on beaches and in time darken, and become weathered and hard. Some shark eggs resemble a pine cone in texture, shape and size. They can float, and are combustible.

Space debris: From time to time, pieces of disintegrating satellites fall to Earth. If they survive the fiery descent through the atmosphere, they may be melted into charred lumps. However, it is fairly unlikely that they would be unrecognisable as products of human technology. It would also be unlikely for them to be dangerously radioactive, unless they had actually housed the satellite's nuclear power source. On the other hand, the origin of items falling from passing aeroplanes would certainly be unmistakable, since they would not be subjected to anywhere near the same level of heat as objects entering Earth's atmosphere from space.

Tar ball: Spills of crude oil from tankers and oil refinery pipelines can result in the formation of tar balls, which are collections of long chain hydrocarbon molecules that can float for extended periods in the sea, eventually being washed up on beaches. They are black, shiny clumps that over time become quite hard. Like all hydrocarbons, they are combustible.

Tektite: A glassy stone that may look like a teardrop, ball, disk, rod, dumbbell or button. It is usually very small, and black, green or amber in colour, with a grooved or pitted surface. It is thought that tektites resulted from comets or asteroids crashing to Earth. This forced ejections of molten material from Earth's crust and the object itself into the atmosphere where it cooled and fell back to Earth. Another cause is as a result of lava masses being hurled high into the atmosphere from volcanoes. Whatever the cause, these liquified rock materials cooled and solidified in the upper atmosphere into glassy stone.

The mystery object - report

Write on the back of this page if you need more space.



List some of the possible materials that the mystery object could be.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

What actually IS the mystery object? _____

What are the reasons for your selection?

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Do you think that the object has any scientific (or other) significance, and if so, what should be done with it?

② The traffic accident

TEACHER'S NOTES

In this case, a bicycle and a motor car collide at an intersection. Both the cyclist and the motorist blame the other party.

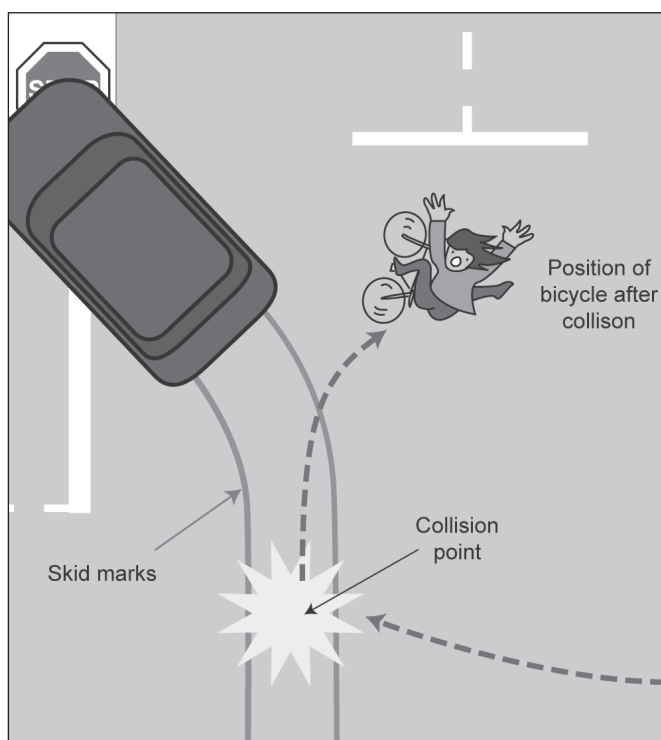
The questions

- What are the possible causes of the accident?
- What is the single most likely cause?
- What is the supporting evidence?

What are the possible causes?

There are usually conflicting views on any issue in which blame is to be apportioned. Each of those directly involved will give a personal slant to the story.

- Some will recount the events with reasonable accuracy.
- Others will unwittingly distort the facts by selectively remembering details of the incident.
- Some will deliberately lie to protect their own interests.



How can the investigators arrive at the truth?

They need to look at the supporting evidence:

- Eye witness descriptions of the accident (these sometimes conflict)
- Expert opinions
- Facts gathered from the scene of the accident

The students' task is to make the most plausible conclusion based on the available evidence. If students have different conclusions, these can be discussed and debated.

DEBATE TOPIC

Anyone in charge of a vehicle using the public road system should have a road user licence, the right to which is regularly reassessed.

Background

Our road system is becoming increasingly congested as more and more people make use of public and private transport (for example, cars, buses, trucks, motorbikes and pushbikes). As well, the power and sophistication of these vehicles is increasing. At the moment, only those in charge of motorised vehicles require a driver's licence, however, they must share the road with many who do not hold a driver's licence.

Gaining a licence ensures a certain level of knowledge and driver skill. Can the same be said of (for example) a cyclist? Or pedestrians? Once a person has a licence, there is at present no requirement to be reassessed, unless a person has had their licence cancelled for some reason. This takes no account of significant changes in age and health, and the impact these can have on skill levels.

On the other hand, there is the issue of driver experience and safety record. Why force a licensed driver who has had a long, blemish-free driving career to be retested?

RELATED ACTIVITIES

Testing reflexes

Just how quick are your reflexes? In fact, they differ from person to person. Obviously, Formula 1 world champion Michael Schumacher has quick reflexes. Yours might be relatively slow compared to his, but is the difference really significant or important? Is the speed of our reflexes a normally distributed characteristic? What other human characteristics are normally distributed – height? weight? intelligence? Can we improve our reflexes through training? Do they change as we grow older?

Devising a way to accurately measure reflexes is a problem in itself. Most people have tried

2 The traffic accident

TEACHER'S NOTES

the 'catch the ruler' test (described below) but the error of measurement is high, so it is not a reliable method. Students will certainly have their own ideas about how to measure reflexes. Whether there are the resources to implement these ideas is another matter. Testing the reflexes of the whole class will give the students an idea of the performance range for general population.

'CATCH THE RULER' TEST

The tester holds a 30-centimetre ruler above the hand of the person whose reflexes are to be tested, with the thumb and index finger poised to grasp the ruler as it is unexpectedly released. The distance the ruler drops before it is caught is a measure of the speed of the person's reflexes. The testee must not move his or her hand or arm in order to catch the ruler – the only movement allowed is the pincer action of the thumb and finger. It is also important that the tester does not signal by any intentional or unintentional personal mannerism when they are going to release the ruler.

ALTERNATE TEST

A little bit of basic computer programming knowledge can result in a good reflex tester. Set the computer to start a timing routine for a random number of seconds after the tester clicks the mouse button. At the end of the timing routine, the word START, for example, appears on the screen. In response, the testee presses any key on the keyboard as quickly as possible. This stops the routine and displays the time elapsed between the word START appearing and the testee's press of the keyboard key.

Calculating distances

Once you have worked out how long it takes the average person to respond to a randomly occurring stimulus, you can get the students to do a few mathematical calculations. For example, given that a motor vehicle is moving at 60 km/h, how far will it travel in an emergency before the average driver can get his or her foot to the brake? How far would it travel if the initial speed was 100 km/h? Or 130 km/h? Of course,

these distances will depend on road conditions, the state of the tyres and the quality of the car's braking system.

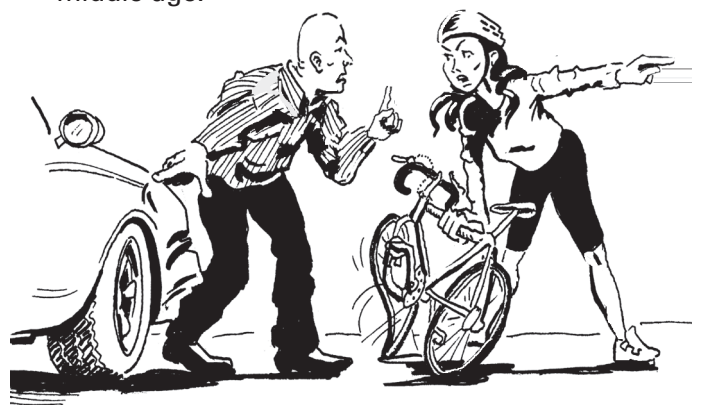
Some stopping distances will be available over the Internet. Typing 'road safety' into the search engine is a good starting point.

Road safety laws

An obvious follow-up question is, should our speeding and licensing laws take account of the lowest common denominator? That is, should road rules be designed so that the person with the worst reflexes can function safely on the roads?

This could lead to a lively discussion on whether or not we should exclude from holding a driving licence anyone whose reflexes fall below the lower bound. It might be argued that a responsible approach to driving, which anticipates and avoids potential accidents, more than compensates for slow reflexes. In fact, this whole topic is a good lead in to the subject of statistics, which are used quite extensively when debating the pros and cons of various road safety issues. For example, is the saying that young motorists are 'most skilled, but most killed' borne out by the facts?

It has also been suggested that older motorists should have more regular licence and medical checks if they wish to continue driving. On what basis has this age group been targeted? For example, are they more accident prone? And what exactly is the age range being suggested by the proponents of this view? Average longevity is increasing, and our standard of community health continues to improve. What was considered to be old age in the early years of the 20th century is now generally regarded as middle age.





The traffic accident

TEACHER'S NOTES

SAMPLE REPORT

Note: The marking scheme is a suggestion only.
You may not want to score the work at all.

Possible causes of the accident (15 points)

- 1 The driver of the motor vehicle was speeding.
- 2 The rider of the bicycle failed to obey the stop sign and so entered the intersection at a potentially dangerous speed.
- 3 The driver of the motor vehicle failed to take sufficient care when entering the intersection.
- 4 The rider of the bicycle failed to take sufficient care when entering the intersection (e.g. did not ride defensively).
- 5 The car was mechanically at fault (brakes, tyres, etc.).
- 6 The bicycle was mechanically at fault (brakes, tyres, etc.).
- 7 Road conditions were poor, and increased the chances of an accident occurring (weather, road surface, general visibility, etc.). Neither party was primarily to blame.

The most likely cause of the accident (15 points)

The cyclist entered the intersection without obeying a stop sign, and so the motorist did not have sufficient time to avoid a collision. The cyclist also had an uncorrected vision problem that may have caused her to miss seeing the motor vehicle, which increased the likelihood of an accident.

Evidence on which your decision is based (15 points)

- 1 One witness stated that the cyclist did NOT stop at the intersection as required by law. The other witness was unable to say whether or not the cyclist stopped at the intersection, but did think that she was travelling too fast immediately prior to the moment of collision.
- 2 Although the cyclist, the bike repair shop owner and two independent witnesses thought that the motorist was travelling too fast, the measurement of the skid marks (8 metres) put the car's speed at 60 km/h (+/- 5 km/h).
- 3 The road conditions were good – a clean, dry bitumen surface.
- 4 The doctor stated that the cyclist had a vision defect requiring correction, and problems with headaches. She also stated that the cyclist was not wearing a safety helmet, as required by law.
- 5 A driver is required to travel at a speed at which he or she can avoid a collision. However, the trees along the service road would have obscured the driver's vision of the cyclist until he was about to enter the main road.
- 6 The force of an impact results from the mass, velocity and direction of vehicles at the point of collision. The bike repair shop owner was not in a position to make an informed judgement about the speed of the motor vehicle. Also, it cannot be assumed that the distance the cyclist was thrown at impact could be put down solely to the speed of the motor vehicle. The angle of collision and the speed of both vehicles would be just two of many factors contributing to this outcome.

The traffic accident

2

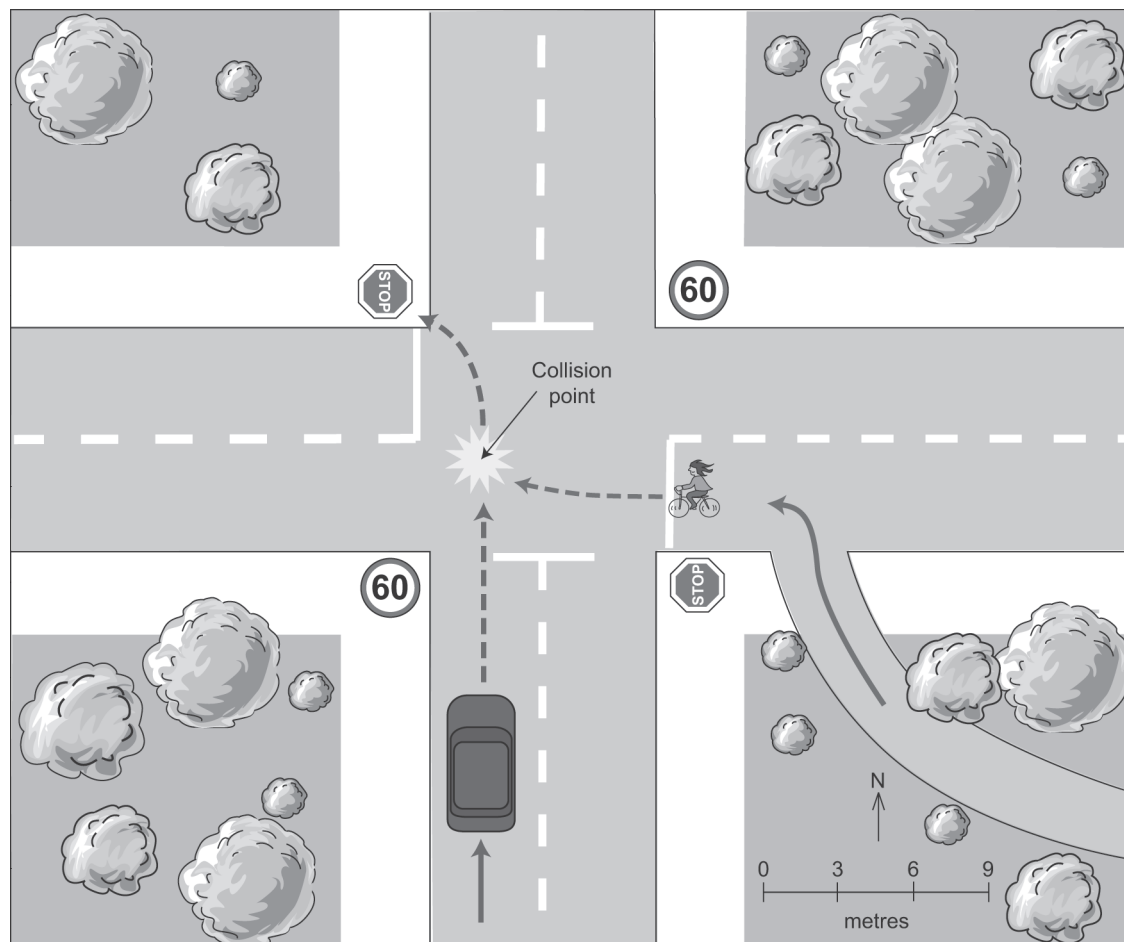
YOUR TASK – You are an investigator for the Australine Insurance Company. Two days ago (Tuesday 17th) an accident occurred involving a bicycle and a motor vehicle insured by Australine. Fortunately, neither the motorist nor the cyclist was seriously injured (the cyclist sustaining only a sprained ankle and minor leg and head injuries). However, the motor vehicle was badly damaged when, after clipping the bicycle, it careered into a street sign. The motorist has put in a claim for the cost of repairing his vehicle. The cyclist has also put in a claim for the replacement of her bicycle and her medical costs, arguing that the motorist was totally to blame for the accident.

This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

- 1 Read your task and study the maps, then write on your report a list of some likely causes of the accident.
- 2 Read the accident reports submitted by the driver of the motor vehicle and the cyclist.
- 3 Distinguish facts from opinions.
- 4 Look for points of agreement among the various reports.
- 5 Determine the single most probable cause of the accident, and list the supporting evidence on which your decision is based.

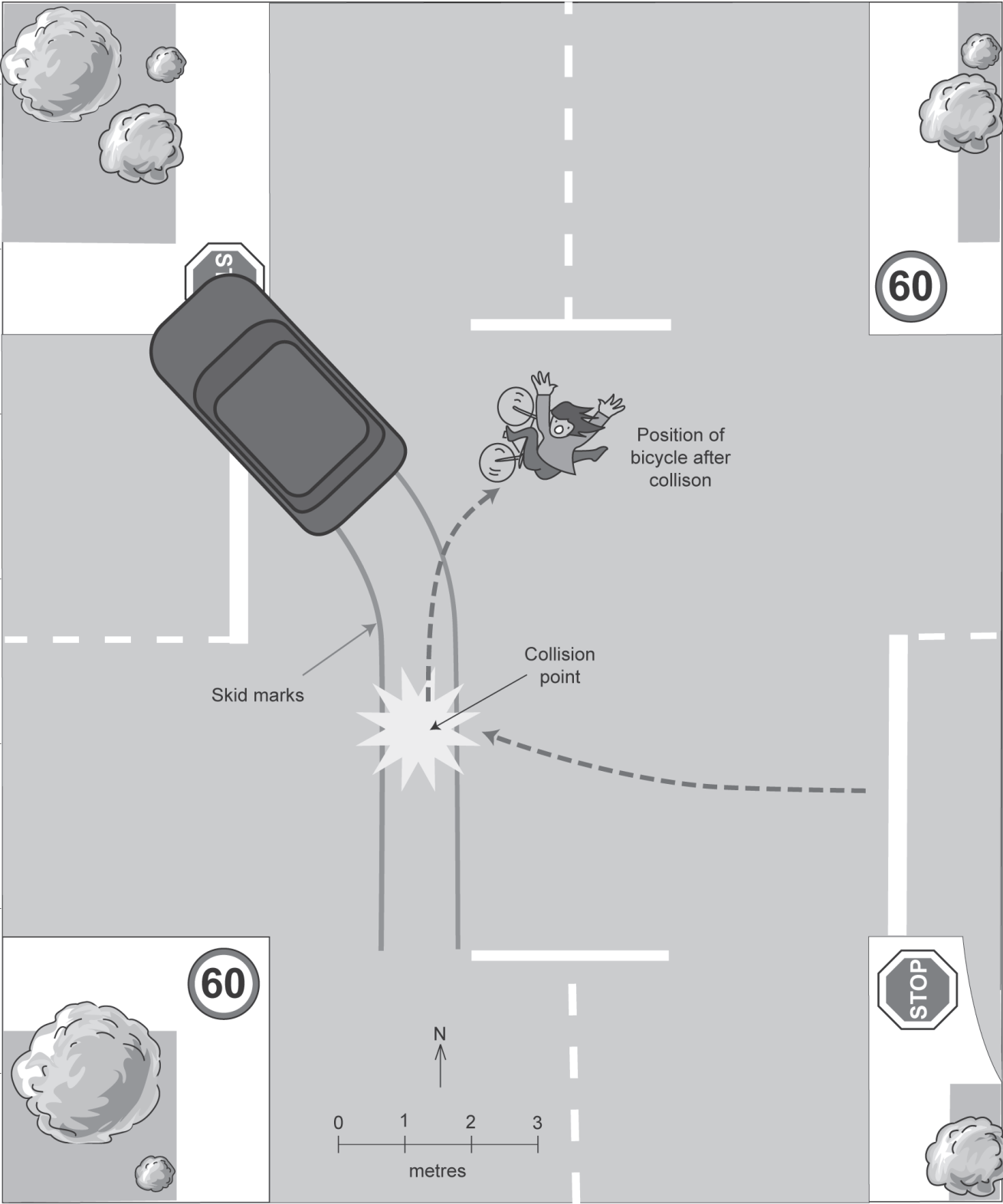
Map of the accident site



The traffic accident



Closer view of the accident site



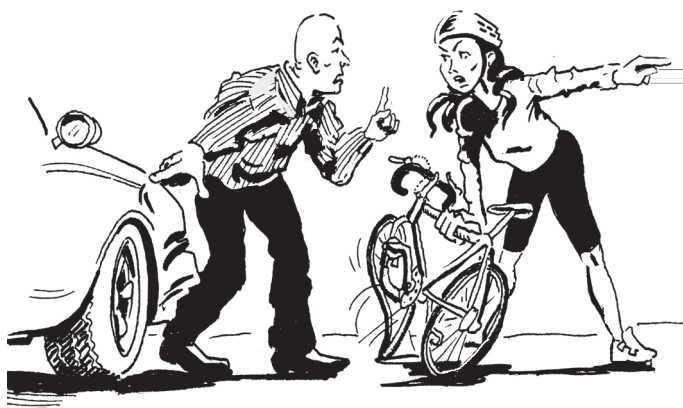
ACCIDENT REPORT – MOTOR VEHICLE DRIVER

I was progressing in a northerly direction at 60 km/h. As I entered the intersection, a cyclist suddenly appeared on my right. The cyclist had failed to obey the stop sign, leaving me little or no chance of avoiding a collision. I applied my brakes forcibly and swerved to the left. Unfortunately I caught the bicycle a glancing blow before veering across the intersection into a stop sign. I quickly got out of my vehicle and hurried to where the bicycle and the rider had come to rest. I was primarily concerned to see that she had not been hurt.

As I approached, she got to her feet and began abusing me. I pointed out that she had caused the accident since she had failed to stop at a controlled intersection.

She continued to abuse me, so I wrote out my personal details and handed them to her, suggesting as I did that perhaps she should see a doctor. She said that she was okay, but it was no thanks to me. As she appeared uninjured, I returned to my vehicle and left the scene.

Although my vehicle had sustained considerable damage to the body panelling, it was still driveable. Later I made a report to the local council concerning the damaged stop sign.



ACCIDENT REPORT – RIDER OF THE BICYCLE

I had just ridden my bike down Service Road from Hilltop Pipes where I work, and after stopping at the corner, entered the intersection from an easterly direction. Suddenly I heard the sound of brakes, and was horrified to see a motor vehicle skidding out of control towards me.

It must have been travelling well in excess of the speed limit, since it had been a fair distance away when I began to cross the intersection a moment or two before.

I swerved to miss the car, but it hit my rear wheel, hurling my bike and me some distance away. I struck my head and twisted my ankle, and was lying dazed on the road when the driver of the motor vehicle came over to me.

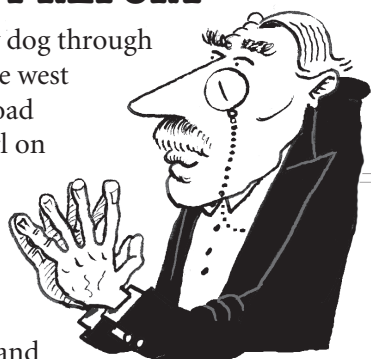
He was very angry, accusing me of causing the accident, and threatening me with police action. His car had extensive body damage, and he said the repair bill was going to cost me a 'pretty penny'. I told him I had hurt my head and twisted my ankle, but all he said was that 'I had brought it on myself'.

Eventually we exchanged personal details, after which I walked my bike back up Service Road to the factory where one of my workmates helped me get home.

My bike was a write-off. Later I went to the doctor to make sure I was okay.

WITNESS 1 REPORT

I was walking my dog through the park along the west side of Service Road when a young girl on a bike raced past me down the hill towards the main road intersection. It's quite a steep hill and is a favourite haunt of skateboard riders.



Anyway, my view of the bike was momentarily obscured by the trees, and I didn't see her again until she flashed into the intersection. I can't tell whether she actually stopped at the corner, but she did seem to be travelling quickly. It was then that I noticed the blue sedan. It was clear that neither of them was going to be able to stop in time. The motorist hit the brakes and swerved to avoid a collision, but all too late. He hit the bike, hurling it and the rider about 5 metres. I thought she'd been killed. Fortunately, she was okay, but she looked pretty sore and sorry. I was a fair distance away, but I could hear them shouting at each other. For a minute it looked like it might end up in a fight. It's hard to tell who was in the wrong, but given the screeching of tyres, maybe the car was travelling too fast.

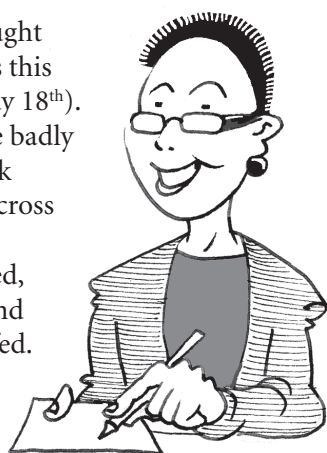
WITNESS 2 REPORT

I was standing at the north-east corner of the intersection, waiting to cross the road. The traffic was light and I was just about to step from the curb when I noticed a bike enter the main street from the Service Road. She had quite a pace up. However, as she reached the intersection she slowed down, but didn't come to a complete stop. She obviously hadn't seen the blue sedan entering the intersection from her left. It was also apparent that the driver of the blue sedan had not seen her either. It was only when the car driver hit the brakes that the cyclist seemed to notice him. They both swerved, but the car clipped the bike, knocking it and its rider to the ground, before ending up against the stop sign. It was a bit of a wreck. The driver got out of his car pretty quickly and raced over to see if the cyclist was okay. Initially he was concerned, but the rider was angry, and in a moment or two they were both accusing each other of being the cause of the accident. Personally, I think the car driver was travelling too fast, although the cyclist should have stopped at the sign. Maybe then she would have seen the car. Surely the onus is on car drivers to look out for others, especially those who are more vulnerable.



BIKE REPAIR SHOP OWNER'S REPORT

Ms Ruby Docic brought me her bike to assess this morning (Wednesday 18th). The two wheels were badly buckled, the rear fork snapped in two, the cross bar twisted out of alignment and dented, a pedal broken off and the leather seat scuffed. The bike is beyond repair and should be replaced. Its



original purchase price was \$865. In my opinion, the car that struck this bike must have been travelling at 70 km/h to 80 km/h in order to cause this much damage.

MEDICAL REPORT



Ms Ruby Docic attended my surgery this evening seeking a physical check-up following an accident in which she was involved at 11.20 am this morning (Tuesday 17th). Ms Docic received mild abrasions to her knees and elbows, and a solid blow to her head that caused contusions and a slight concussion. She also had

strained right ankle ligaments. Unfortunately Ms Docic had not been wearing her safety helmet at the time. She has complained of headaches for several months, and testing has shown that she will require glasses to overcome a vision defect. I understand she was on her way to pick up her new glasses when the accident occurred. It is most unlikely that Ms Docic will suffer any long-term ill-effects from the accident, and I expect her to be fit to return to work by Thursday 19th.



ROAD RULES

- 1 A driver shall enter and pass through any intersection at such a speed that they will be able to stop their vehicle in time to avoid a collision with any person or any other vehicle.
- 2 When approaching or entering an intersection, a driver must give way to any other vehicle approaching or entering the intersection from a separate carriageway on their right except when a vehicle on the right faces a ‘Give Way’ sign or a ‘Stop’ sign.
- 3 Where a ‘Stop’ sign is erected to face drivers approaching an intersection, a driver who has passed or is passing the sign:
 - a shall stop their vehicle before reaching and as near as practicable to the stop line
 - b may then proceed but shall give way to every vehicle travelling along or turning from any intersecting carriageway.

FOUR-DAY WEATHER DETAILS

Saturday 14 th	Fine and sunny. Light north-easterly winds. Top temperature 30°C
Sunday 15 th	Fine and sunny. Light north-easterly winds. Late change, winds swinging to the south. Heavy showers. Top temperature 25°C
Monday 16 th	Early rain clearing to a fine day. Light south-easterly winds. Top temperature 27°C
Tuesday 17 th	Fine and sunny. Light east to north-easterly winds. Top temperature 31°C

RELATING LENGTH OF SKID MARKS TO VEHICLE SPEED

Calculations are made on the basis of a clean, dry, bitumen road.

Length of skid marks	Estimated speed
12 metres	80 km/h
10 metres	70 km/h
8 metres	60 km/h
6 metres	50 km/h
5 metres	40 km/h

Note: These figures are estimates, and may only be regarded as accurate to + or – 5 km/h. If the road is wet or greasy, then the condition of the motor vehicle’s tyres must also be taken into consideration. It is also assumed that the motor vehicle does not have an anti-lock braking system (ABS).

The traffic accident – report

Write on the back of this page if you need more space.



Possible causes of the accident

[illegible]

The most likely cause of the accident

Evidence on which your decision is based

[illegible]

Estimated motor vehicle speed _____ km/h



The school excursion

TEACHER'S NOTES

In this mystery, several classes from the local school go on an end-of-year excursion to the neighbourhood recreation reserve, a traditional outing that has gone without incident for a number of years. On this occasion, however, some of the students become ill during the afternoon and have to be taken to hospital.

The questions

- What are the possible causes of their illness?
- What is the single most likely cause?
- What is the supporting evidence?

What are the possible causes?

- Is it something they ate or drank?
- Did they participate in some activity that exposed them to a health risk?
- Were they simply out in the sun too long?
- Were they coincidentally all coming down with some viral infection that just happened to manifest itself on the day of the picnic?

What is the most likely cause and what is the supporting evidence?

The students undertaking this case are required to assess the available information and reach an informed decision on this question.

Note: The figures provided in the toxicity chart are not intended to be scientifically accurate, but have been arbitrarily created for the purpose of the exercise. To reduce the chance of potential misunderstanding with real figures, no units have been included.

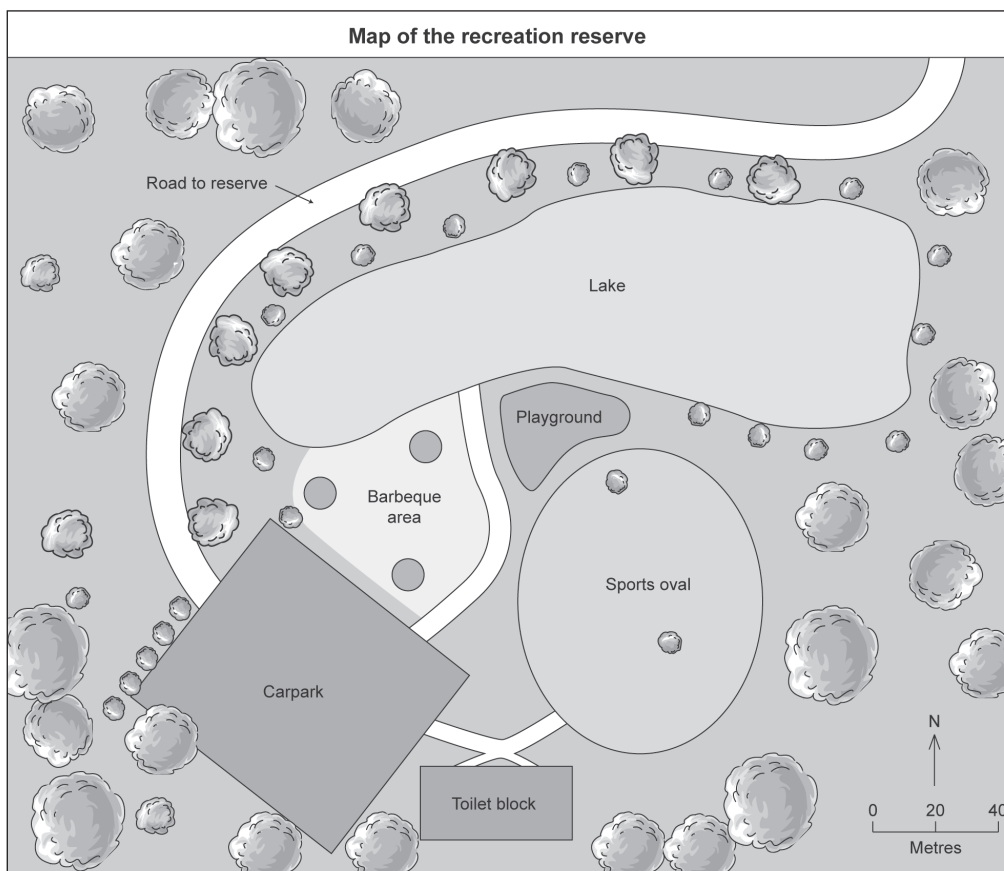
RELATED ACTIVITIES

This activity requires students to apply logical thinking and scientific methods of analysis to a problem. However, it can also be a lead-in to a range of other health and science topics. The cause of the mystery illness is either something the picnickers have eaten or drunk, a poisonous substance (such as weed killer or exposure to algal toxins) that has been absorbed through their skin, an allergic reaction to insect stings or plant nettles, or dehydration, sunburn, and/or heatstroke.

Discussion subjects

- The prevention and cure of the common types of ailments and afflictions listed above.
- The use of pesticides and alternative pest control strategies. (Do the ends justify the means?)

- The preparation and storage of food (for example, cooking, reheating, refrigerating, bottling, pickling and smoking). Students should attempt to draw up some simple guidelines in regard to this issue.
- Micro-organisms
- Algal blooms
- Weathering the weather; for example, avoiding hypothermia, dehydration, sunburn and heatstroke





The school excursion

TEACHER'S NOTES

There are many interesting experiments related to micro-organisms. For instance, students can study the effect of temperature on food, for example, comparing milk stored at room temperature with milk in the refrigerator. How do the results obtained match against the use-by dates on long-life milk cartons? What exactly do 'use-by' dates mean? Why do unopened canned goods have a longer shelf life?

Try this: Gently boil a small amount of milk in a science flask, then quickly and carefully seal the flask and place it on the window ledge of the classroom. Does the boiled milk last longer than a similar sample of unboiled milk, or milk stored within a refrigerator?

RESOURCES

The school library probably has many books about health and hygiene, and there are of course, many official Government Health Department websites.

SAMPLE REPORT

Note: The marking scheme is a suggestion only. You may not want to score the work at all.

Possible causes of the illness (15 points)

- 1 They may have suffered sunburn/sunstroke.
- 2 Some may have got food poisoning from either the food or the drink provided on the excursion.
- 3 Some may have swum in the algae-infested section of the lake.
- 4 Some may have suffered allergic reactions to insect bites or stings.
- 5 They may have adversely reacted to weed killer from the recently sprayed grass.

Cause of illness and estimated time off school (10 points)

Eight of the nine students admitted to hospital had drunk the cordial. Since bottles of the type used for the cordial prepared prior to the excursion were found to contain no contaminants after having been washed, only those who drank the cordial prepared from the tank water got sick. This cordial was found

to have a dangerously high level of bacterial contamination. None of the food was found to be contaminated, and although a section of the lake had a dangerous toxic algal bloom, no one who swam in the lake was admitted to hospital. The only other plausible explanation for the illness of the ninth student was that he suffered sunburn/sunstroke. Most, if not all, of those who were sick would probably be back at school within 14 days.

Evidence (25 points)

- 1 None of the pre-prepared food was found to have bacterial contamination.
- 2 The toxicity level of the residue of the weed killer used to spray broad-leaf weeds on the oval was within safety limits.
- 3 Although a section of the lake had a dangerous toxic algal bloom, no one who swam in the lake was admitted to hospital.
- 4 Eight of the nine students admitted to hospital drank the cordial. Since bottles of the type used for the cordial prepared prior to the excursion were found to contain no contaminants after having been washed, it seems likely that only those who drank the cordial prepared from the tank water got sick. This cordial was found to have a dangerously high level of bacterial and faecal contamination, caused by the fact that it was collected from the toilet block roof upon which there were probably possum droppings. On this occasion, a decomposing possum carcass was also found in the spouting that fed the storage tank from which the water was taken to make the second batch of cordial.
- 5 The ninth student admitted to hospital (Terry Nesbit) had played soccer on the oval and had probably suffered severe sunburn/sunstroke as a result (see the Doctor's report and Colin Forbes' statement). However, he had not drunk any cordial. Four others had also played soccer, but had also drunk the cordial. The remaining two had not played soccer nor had sunburn, but had drunk the cordial.



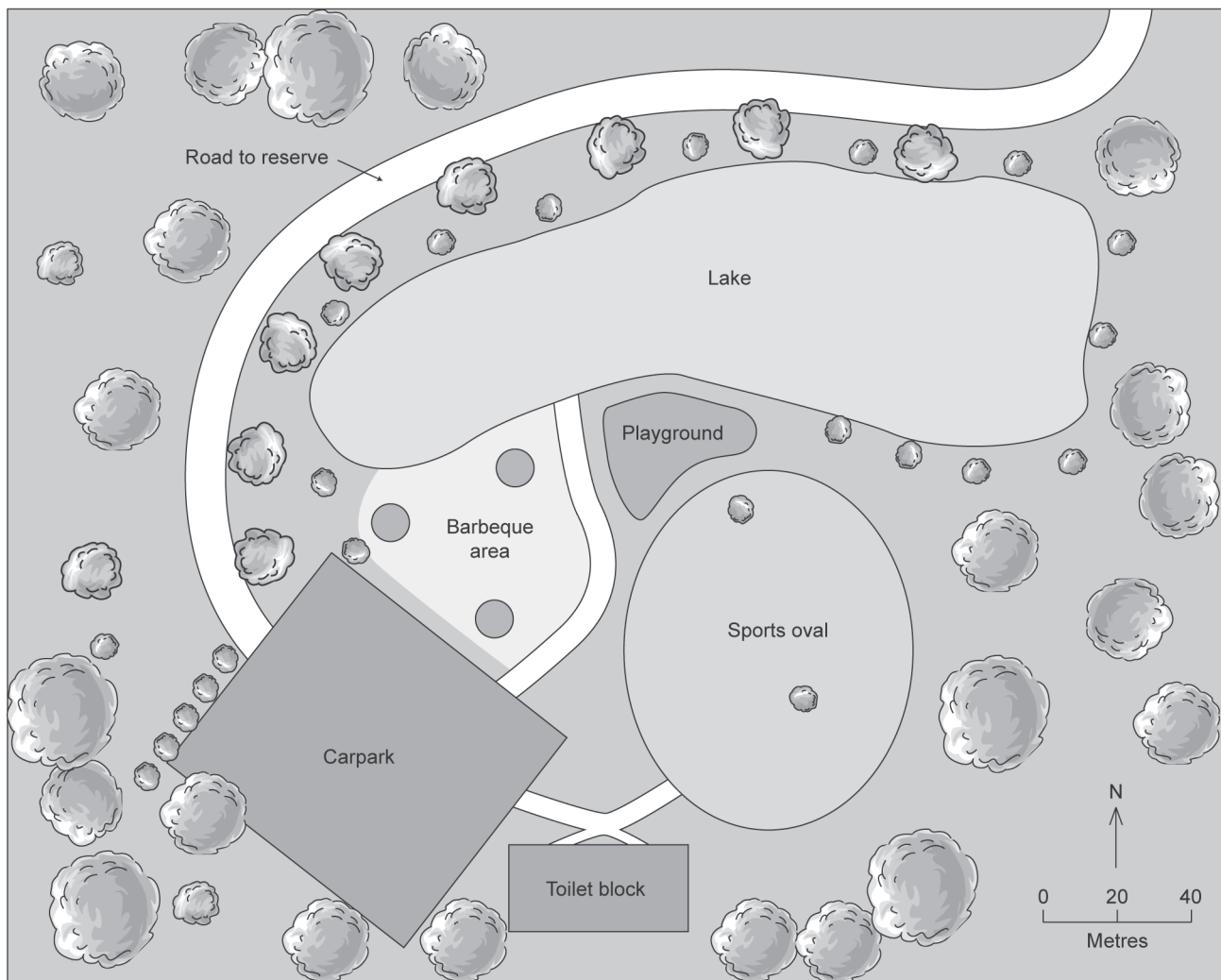
YOUR TASK – You are the principal of Sunnyvale Secondary College. Two days ago (Tuesday 17th), several classes had their traditional excursion to the local recreation reserve. During the afternoon, nine of the 40 students attending the excursion became ill and were rushed to hospital. All nine were treated in the hospital's casualty section, after which they were admitted as in-patients, where most of them still remain. You must discover how these students became ill.

This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

- 1 Your first task is to speculate on some of the more likely alternative causes of the students' illness.
- 2 Use the information that follows to decide which of these alternatives is the most probable cause.
- 3 Look for relationships between the various activity lists and reports.
- 4 Indicate how long you think these students are likely to be absent from school.
- 5 Your report should include a list of the supporting evidence on which it is based.

Map of the recreation reserve



STUDENT STATUS

Admitted to hospital:

Christine Baker
 Danny D'Angelo
 Alan Edwards
 Colin Forbes
 Michael Kostos
 Terry Nesbit
 Mary Panopolis
 Brenda Smith
 Kyja Sokim

Ate the barbequed sausages:

Carl Considine
 Alan Edwards
 Colin Forbes
 Guy Gregory
 Margaret Hughes
 Helen Jeffreys
 Lisa Leung
 Walter Matthews
 Terry Nesbit
 David Nolan
 Ray Reagan
 Maria Sentic
 Brenda Smith
 Jimmy Solway
 Peter Wong
 Jiro Yokoyama

Swam in the lake:

Frank Burgess
 Barry Charles
 Carl Considine
 Luana Okaro
 Edward Jones
 Lisa Leung
 Api Rupuma
 Ray Reagan
 Alex Santinia

Played soccer on the oval:

Graham Arthur
 Danny D'Angelo
 Alan Edwards
 Colin Forbes
 Michael Kostos
 William Moon
 Terry Nesbit
 Danny Phung
 Jiro Yokoyama

Needed to be treated for sunburn:

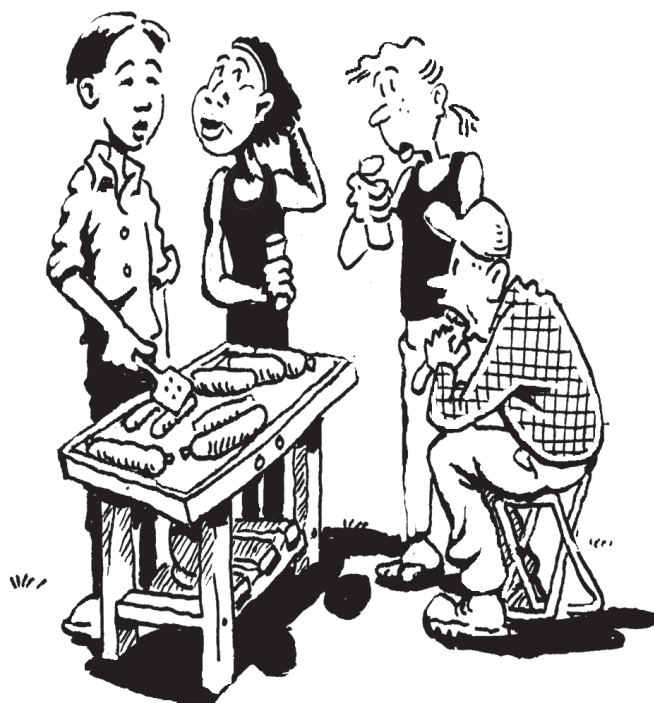
Graham Arthur
 Jenny Cook
 Danny D'Angelo
 Barry Gilbert
 Michael Kostos
 Terry Nesbit
 Danny Phung
 Api Rupuma
 Phillip Thomas

Ate the chicken sandwiches:

Christine Baker
 Barry Charles
 Danny D'Angelo
 Demir Fikret
 Michael Kostos
 Jackson Lane
 Donna Malone
 Adrian Morris
 Anne Rogers
 Kyja Sokim
 Phillip Thomas
 Paul Williams

Drank the cordial:

Graham Arthur
 Christine Baker
 Barry Charles
 Carl Considine
 Danny D'Angelo
 Alan Edwards
 Colin Forbes
 Guy Gregory
 Margaret Hughes
 Helen Jeffreys
 Michael Kostos
 Lisa Leung
 Donna Malone
 William Moon
 Andrew Nicholson
 David Nolan
 Mary Panopolis
 Api Rupuma
 Ray Reagan
 Maria Sentic
 Brenda Smith
 Kyja Sokim
 Jimmy Solway
 Benji Tan
 Phillip Thomas
 Peter Wong
 Paul Williams
 Jiro Yokoyama



RESULTS OF TESTING

Test	Result
On the chicken sandwiches	The level of Salmonella contamination in the chicken sandwiches was 15.
On the barbeque sausages	The barbeque sausages were found to be free of any bacterial contaminants. However, they did contain an unusually large amount of preservative which might cause gastric irritation if more than 15 sausages were consumed by a person within a 60-minute period.
On the cordial	No chemical contaminants were found, however the sample of the cordial tested had a bacterial contamination level of 35.
On the lake water	• The toxicity levels resulting from the algal bloom ranged from 5 to 45 depending on where in the lake the water samples were drawn. • Copper sulphate concentration levels were 5.
On grass from the sports oval	The concentration level of 245-T was in the range of 3 to 5.
On the water samples from the toilet block	The water from the hand basin taps contained a noticeable but safe level of iron oxide (rust). The tank water – although clear to the eye – contained some faecal material, and had a bacterial contamination level of 35 as a result of a decomposing possum carcass in the toilet block spouting.
On the plastic bottles	Several plastic bottles of the type normally containing laundry detergent were checked at random for residual contaminants after they had gone through the washing process. No contaminants were detected.

OTHER REPORTS

Doctor’s report

Nine students from the Sunnyvale Secondary College were admitted to hospital. Most exhibited varying degrees of the following symptoms: acute abdominal pain, vomiting, diarrhoea, headaches and slightly elevated temperatures. Some of them were also suffering moderate to severe sunburn and dehydration.

Details of an interview with Mr William Mifsud (Chemistry teacher)

PRINCIPAL Had the bottles used for the cordial previously contained laundry detergent?

MIFSUD Yes, they had. However, the bottles had been thoroughly washed and would have been free of any dangerous contaminants.

PRINCIPAL Have you previously been responsible for preparing the cordial for the school picnic?

MIFSUD Yes, for the last five years.

PRINCIPAL Did you follow the same procedures on each of those occasions?

MIFSUD Yes.

PRINCIPAL To your knowledge, was anyone ever ill on any of those previous occasions?

MIFSUD No.





Statement from Colin Forbes (student)

I spent the early part of the morning helping Mr Mifsud unload the food and drink from the school bus. We then had a game of soccer on the oval, which Michael refereed. We divided alphabetically. It was Shirts against Skins, and I was one of the Shirts. Unfortunately, a number of the Skins forgot to use sunscreen and got a bit sunburnt.

Later, I helped run the barbeque. The sausages weren't bad. I ate six in about 15 minutes. They were a bit salty or something, so I washed them down with a couple of large glasses of cordial.

After lunch someone suggested having a swim in the lake. It looked a bit dirty in some spots but we found a suitable area. Later, someone found a sign warning us against swimming. But it seemed all right to me.

Anyway, about mid-afternoon I suddenly began to feel really sick. I had pains in the stomach and felt nauseous. In no time at all, a number of others felt sick too. Mr Wing had to drive us all to the hospital in the school bus. He thought I might have had too much sun, but I think I'd been careful about this.

PRINCIPAL'S NOTES

Ants and bees	It was stated by several students that ants and bees had been a problem in the barbeque area but no one reported being stung, and none of the students is known to be sensitive to insect stings.
The cordial	The cordial was pre-prepared at the school by William Mifsud, the school chemistry teacher, and brought to the recreation reserve in large plastic bottles of the type normally used to store laundry detergent. Some students claimed that this cordial had a noticeable chemical taste. During the day, the cordial ran out and a new batch was made up using water from a tank attached to the roof of the toilet block. It was not possible to test the original supply of cordial as it had all been consumed, so the remnants of the second batch were submitted for testing. Those who drank the cordial were unaware of which batch they were consuming.
The lake	There was evidence of a toxic algal bloom in some parts of the lake, prompting the City Health Officer to suggest that the lake should not be used for swimming. The Council was having the lake treated with copper sulphate to control the algae, and then planned to increase the water flow-through rate in order to prevent any future occurrences.
Grass on the sports oval	A sign near the sports oval indicated that it had been recently sprayed with a broad-leaf weed killer, and warned of the possibility of allergic reactions.
The toilet block	Hand basins were connected to the mains supply water, but old pipes made the water very rusty. Soap and paper towel dispensers were located adjacent to each hand-basin. A 5000-litre water tank provided what seemed to be a much clearer source of water. At some stage during the day, all students used the toilets.
Preconditions	Several days later, I interviewed the students who attended the picnic. They all suggested that no precondition existed that could have given rise to the sudden illness of the nine who were hospitalised.



FOUR-DAY WEATHER DETAILS

Saturday 14 th	Fine and sunny. Light north-easterly winds. Top temperature 30°C. UV rating – very high to extreme.
Sunday 15 th	Fine and sunny. Light north-easterly winds. Late change, winds swinging to the south. Heavy showers. Top temperature 25°C. UV rating – high (early).
Monday 16 th	Early rain clearing to a fine day. Light south-easterly winds. Top temperature 27°C. UV rating – very high.
Tuesday 17 th	Overnight heavy rain clearing in the morning to a fine and sunny day. Light east to north-easterly winds. Top temperature 33°C. UV rating – extreme.



TOXICITY CHART – MAXIMUM CONTAMINANT SAFETY LEVELS

Contaminant	Concentration	Symptoms or effects
E coli	20	Gastric disorders, diarrhoea, headaches, vomiting. Illness can be of 2–14 days duration.
Salmonella	20	Gastric disorders, diarrhoea, headaches, vomiting. Illness can be of 2–14 days duration.
245-T	30	Headaches, vomiting, abdominal pain. 2–7 days recovery period.
Blue-green algae	40	Headaches, vomiting, abdominal pain. High concentrations can cause death.
Phosphorus (in the form found in most laundry detergents)	7	Headaches, vomiting, abdominal pain. 5 days recovery period.
Copper sulphate	25	Headaches, vomiting, abdominal pain. High concentrations can cause death.

- Notes
- 1 The figures provided in this toxicity chart are not intended to be scientifically accurate, but have been arbitrarily created for the purpose of the exercise.
- 2 The severity of the health effects of these contaminants will depend on the concentration levels and the quantities consumed, and may vary from individual to individual.

The school excursion – report

Write on the back of this page if you need more space.



Possible causes of the illness

[illegible]

Cause of illness and estimated time off school

Evidence on which your decision was based

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



In this mystery, a UFO (unidentified flying object) is seen over the northern suburbs of the city during the early hours of the morning. It is witnessed by a few people out and about at that hour.

The question is, what is it?

- Is it an extra-terrestrial spacecraft?
- Is there a more down-to-earth explanation?

The support material includes:

- eye-witness accounts
- comments from authorities
- views from local residents, appearing in the Letters to the Editor section of the local newspaper, *The Metropolis Times*.

This activity does not attempt to resolve the broader issue of the existence of UFOs. However, there is at least one plausible solution to this particular mystery that can be deduced from the student material.

Few topics capture people's imaginations to the same degree as UFOs. In the minds of many, the UFO is synonymous with flying saucers and intelligent extra-terrestrials (alien creatures). Science fiction abounds with stories about UFOs and ETs. Many depict the aliens as hideous and warlike, for example, H.G. Well's *The War of the Worlds*, while others have a much more benevolent view, such as Spielberg's *ET*, and television's *My Favourite Martian*.

This topic is full of research and discussion opportunities for students, who should:

- distinguish fact from fiction
- look at how science relates to the many eyewitness accounts of UFOs and the apparent abundance of photographic evidence. Just how binding are the laws of physics?
- consider the argument put forth by UFO supporters that to dismiss ETs on the basis of our current understanding of science and technology constraints on interstellar and intergalactic travel is to ignore how far humanity has developed in the last 500 years. Just because it seems impossible to us, it doesn't mean that it's altogether impossible.

DEBATE TOPIC: UFOs are just a figment of the imagination.

BACKGROUND

Interest in UFOs gained impetus in the early 1950s. Some believed that we had already been visited by alien spacecraft, as evidenced by the large number of unofficial UFO sightings that occurred around the world during that time.

One popular view was that aliens were merely keeping Earth under surveillance, given the violent history of its dominant species, and more recent rapid advances in technology. Others were of the opinion that contact with intelligent extra-terrestrials had already been made, but that governments had engaged in a conspiracy of concealment.

Others questioned the logic of such sightings, suggesting that the mind-bogglingly huge distances involved in interstellar and intergalactic travel made such claims untenable. A round trip might take hundreds of thousands of years. However, modern science fiction writers have created new hope for believers through their introduction of seemingly plausible concepts such as worm holes, warp drives and hyperspace.

One more idea to consider – would NASA be spending billions of dollars trying to establish whether the building blocks of life exist elsewhere in the near universe, if governments were concealing the arrival of intelligent aliens?

RELATED TOPICS

English astronomer Sir James Jeans suggested that there are more stars in the universe than grains of sand on all the beaches on Earth (approximately 10^{27}). This being true, it seems reasonable to assume that at least somewhere in the universe there will be an intelligent life form, which has the scientific and technological capacity to travel 'to the stars.' Would it then be fair to say that Earth might have been – or might yet be – visited by some of these intelligent extra-terrestrials?

The analysis of this question enables students to apply fascinating mathematics and science concepts. For a start, there is the question of the probability of there being other life forms in the universe.

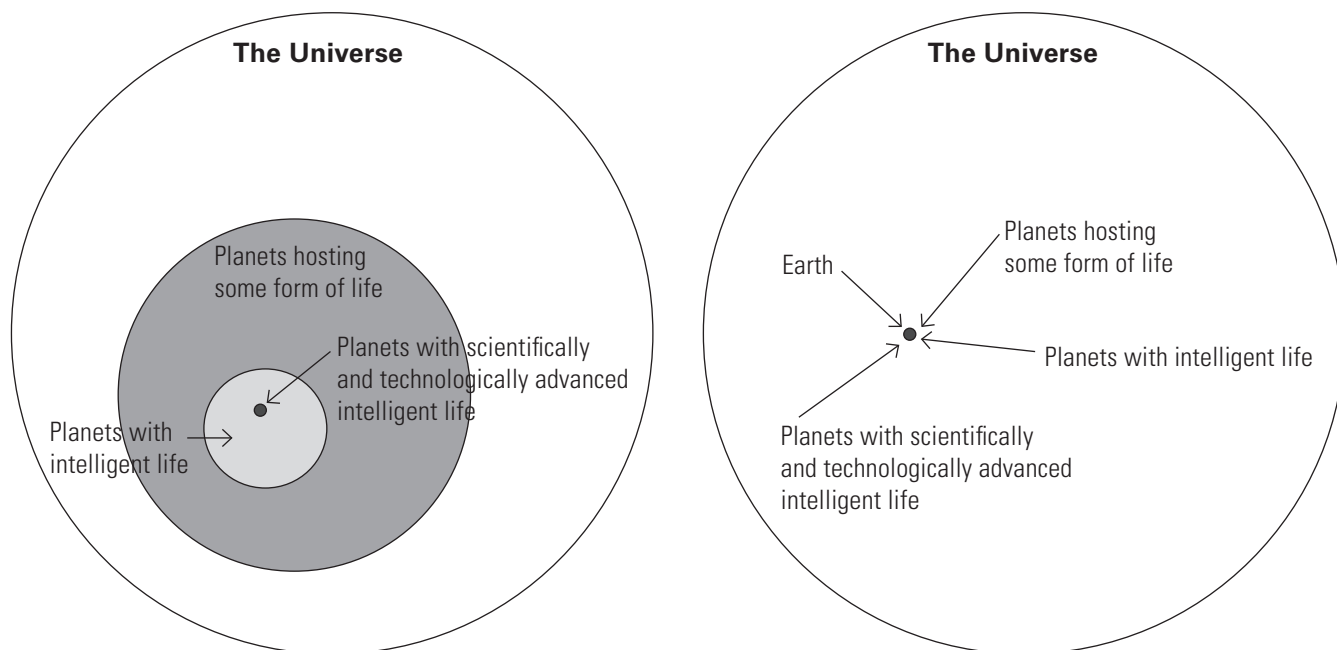


Northern lights

TEACHER'S NOTES

NUMBER OF STARS IN THE UNIVERSE

The author is not sure why anyone would have taken the time to work this out, but students may enjoy attempting this estimation exercise. It's tedious, but interesting. Discuss different methods by which students could work out the number of grains of sand in (say) 50 cubic centimetres. One way is to use water displacement. Add enough sand to increase the volume of a measuring bottle by 50 cubic centimetres, then share out the sand and count the grains. After that, it's just a matter of multiplication – lots of multiplication!



Obviously, the domains shown above are completely out of proportion. You could discuss their possible relative sizes. Even if we are not alone, the likelihood that our planet is the 'chosen one' seems remote. By way of comparison, it's true that someone nearly always wins Tattsлото but the chance of it ever being one particular individual is minimal.

*Then there's the question of **distance**. Students will need to do a bit of research for this activity. How big is our solar system? What is the distance to our nearest star? What is the size of the Milky Way Galaxy? How far away is the next closest galaxy to our own? How do we measure these distances? How long would it take to get to these places travelling at (say) the top speed of a modern jet fighter, or the speed of the Apollo rocket that took astronauts to the Moon? What about travelling at the speed of light? Is it possible to travel at such a speed? What conclusions can one draw?*

*Finally, we have to consider **motivation**. In 1969, the United States sent a spaceship to the Moon. By around 2025, it is hoped that a human will have set foot on the planet Mars through the efforts of the US space program. When making this recent decision, the US President committed the American people to an expenditure of many tens of billions of dollars. The event will not occur within his own period of office, maybe not even within his own lifetime. What if the journey was going to take a hundred lifetimes, and that the astronauts might never return? Would any government ever make the enormous commitment necessary for such a venture, given that they would almost certainly never know the outcome? Reference could be made to early Earth explorers such as Columbus.*

Use ITC to help do space-related calculations

Undertaking the sorts of calculations mentioned here can be time-consuming and tedious, even with an electronic calculator. Students might like to try developing a spreadsheet program to do the calculations. Enter the distance in kilometres in one cell, and the speed of the space traveller in km/h in another, and get the program to do the work. Similarly, for distances in light years, get the program to do the conversion to kilometres.



Northern lights

TEACHER'S NOTES

CHECK THESE OUT

Is it likely that we have ever been visited by intelligent extra-terrestrials? Maybe all they needed was warp speed or a convenient worm hole.

These websites could be used to check out some hypotheses:

- A web site that includes photographs of possible UFOs
– <http://vzone.virgin.net/darryl.smith/ufo/crowdedskies.html>
- Was Roswell just a big hoax? You be the judge!
– www.unmuseum.org/fairies.htm
- Are there any UFOs where you live?
www.geocities.com/Area51/Zone/2788/

TRY THIS ACTIVITY

Ask students to describe and illustrate an ET and they'll probably give you the standard Spielberg stereotype, or else something even more weird and totally over the top (multiple heads, legs and antennae, grotesque bodies, and so on). However, you can make this description more challenging.

Begin by discussing what scientists mean when they talk about the basic concept of 'life'. What are the essential characteristics of a living thing? (Nutrition, respiration, reproduction, and so on) Perhaps give them a rudimentary introduction to Darwinism and the notion of environmental adaptation.

Finally, get students to choose a planet like Mars or Venus and research the conditions on that planet. Once they've got the necessary background information, have them create an ET that is truly adapted to its environment, taking such things into account as the food chain and the interaction of species.

CONSIDER THIS

Stephen Hawking's Universe by David Filkin,
BBC Books, London, 1997

SAMPLE REPORT

Note: The marking scheme is a suggestion only. You may not want to score the work at all.

Possible causes of the UFO sighting (15 points)

- 1 An alien spacecraft (for example, flying saucer)
- 2 A commercial or military aeroplane
- 3 A weather balloon
- 4 Lightning (forked, sheet or ball)
- 5 A shooting star (meteor)
- 6 A trick of the imagination
- 7 An optical illusion

Most plausible explanation for the UFO sighting (15 points)

A team of four skydivers, led by Grant Wilson, carried out the illuminated, sound-enhanced skydiving 'stunt' over the northern suburbs of the city with the deliberate intention of hoaxing the public into believing that it was a UFO.

Reasons (20 points)

- 1 Official weather reports tend to exclude lightning as a cause, and two of the eyewitnesses were convinced that there wasn't any lightning.
- 2 Official spokespeople discounted the possibility of the UFO being either a military or commercial aeroplane.
- 3 A private aircraft hired by Grant Wilson was reported to have been flying in the area at the time. Its destination was logged as Kingston, the location of a forthcoming skydiving competition.
- 4 Grant Wilson, skydiver and reputed 'wag' and sceptic, had suggested in the 'Letters to the Editor' column of *The Metropolis Times* that thick-headed and gullible UFO devotees needed to be taught a lesson.
- 5 The Grant Wilson-led team of four skydivers were famous for their 'sound and light' show.

YOUR TASK - You are a reporter for *The Metropolis Times*. Two days ago your office received numerous reports of an unidentified flying object sighted over the northern suburbs of the city during the early hours of the morning. The afternoon edition of the newspaper included a short item on these 'so-called' eyewitness reports, but your editor now wants to know whether the subject warrants a follow-up article. So far, no one has managed to provide an authoritative explanation for the sightings, although there are lots of different theories floating around. Your job is to investigate the issue more thoroughly in order to discover the real identity of the UFO. Was it from outer space? Was it the product of extra-terrestrial intelligence? Was it of a sinister origin, or did it have a more mundane, down-to-earth explanation?

This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

- 1 Read your task and think about the sort of evidence that might help you determine what the UFO might be.
- 2 Make a list of plausible alternative explanations.
- 3 Read the eye-witness accounts, the statements from the representatives of the various authorities, and the newspaper articles, and consider whether the information they provide favours any of these alternative explanations. Do they suggest any additional possibilities?
- 4 Determine what you believe to be the most likely cause of the UFO sightings, and provide the reasons that led you to this conclusion.

The Metropolis Times

Sunday 15 July

Northern Lights

Hundreds of worried northern suburbs residents were awoken in the early hours of this morning to a strange light and sound show in the sky above their homes. 'It looked like a huge fire ball zooming across the sky at high speed,' said Mrs Zigkowsky, long-term resident of the area. 'It literally shook me out of my bed.' Frank Waterpond was on his way home from work when he saw the UFO. 'It was like a shooting star zig-zagging back and forth across the sky,' he said.



An unnamed government source stated that it was probably the result of some unusual weather pattern over the city during the last

few days, producing a few random bolts of lightning. However, 12-year-old Joe Panelli, just starting his morning paper delivery round the time of the sightings, said it was like no bolt of lightning he'd ever seen before.



Are we being visited by ETs?

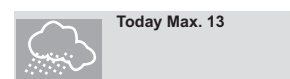
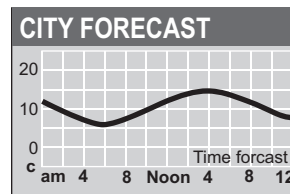
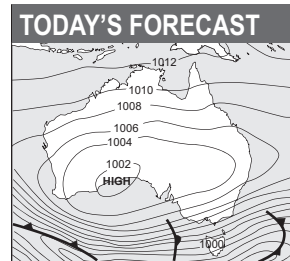
A military media officer said that as far as he knew, there were no aircraft in the area at the time.

The last word went to Mrs Molly Muggleweed, a well-known member of the 'UFOs Really Do Exist' organisation, who said, 'I've always said that they are out there watching us.'

What's your view?

Ring our 50-50 hotline on
161 399

The Weather



Outlook
Tomorrow: Max 15 Few showers.
Tuesday: Max 15 Shower or two.
Wednesday: Max 15 Shower or two.

EYE-WITNESS ACCOUNTS

Mrs Zigkowsky **84-year-old pensioner**

I'd been asleep for several hours, when I was awoken with a great start by a booming sound that seemed to come from right over my house. It literally shook me out of my bed. The room should have been dark, but instead, it was lit up by a bright light shining through my bedroom window. I put on my robe and went to the window and looked out. At first I was dazzled by the light, but after a moment or two I could see a bright object moving about in the sky. Then all of a sudden it was gone. I found it difficult to go back to sleep, wondering if I had just imagined it all.



Joe Panelli **12-year-old student and newspaper delivery boy**

I was just wheeling my bike to the front gate on my way to the newspaper shop to pick up my papers for delivery, when I saw several bright lights in the sky. They seemed to pulse on and off for several minutes. Then there was a loud booming noise, sort of like a distant clap of thunder, but the lights themselves were like no lightning I've ever seen. Mum suggested it might have been a flying saucer, but it didn't look like it to me.



Frank Waterpond **43-year-old night-shift worker**

I'd just put the car in the garage and was walking towards the front door when all of a sudden I noticed a bright light zig-zagging across the sky. At first I thought it was an aeroplane or a helicopter, but then I thought it might be lightning, or something. It didn't last very long. I'm a welder by trade, and maybe a couple of welding flashes had made me see stars for a while. It wouldn't be the first time that's happened. You've really got to be careful when you're using a welder.



Molly Muggleweed **Secretary, 'UFOs Do Exist' Society of Australia**

This is another example of a government cover-up. We've known for decades that these extra-terrestrial spacecraft do exist. Who really built the pyramids, and what about those crop circles? Surely you must agree that 'Chariots of the Gods' exposed the absolute hypocrisy of all those 'head in the sand' doubters. They're putting these latest reports down to lightning. What rubbish! I checked with the Weather Bureau and they said that there was no lightning over the area on Tuesday morning. Believe me, it was a UFO. I've always said they were out there watching us.



EYE-WITNESS ACCOUNTS

A military spokesperson



Our radar picked up an unidentified flying object over the northern area of the city during the early hours of Sunday morning, the 15th. It remained on our screens for about two or three minutes, during which time its behaviour could best be described as erratic. As far as I am aware, there were no military aircraft in the area at the time of the reports. However, we have a logged flight plan of a civilian aircraft in the vicinity during that period. My unofficial opinion at this stage, based on meteorological data, is that the reported lights were the result of sheet lightning.

A government source

We have had several reports of some bright lights over the northern suburbs during the early hours of this morning. It was probably due to some unusual weather we've been having in the last few days. High humidity coupled with a cold upper-air stream is a sure recipe for a thunderstorm. Maybe we just had a few random bolts of lightning. I can assure the public that there is absolutely no evidence to support the ridiculous view that the lights were caused by some sort of extra-terrestrial spacecraft.



A spokesperson for Aircraft Control Centre Metropolis Airport



A small twin-engine aircraft, hired by a Mr Grant Wilson, departed from Metropolis Airport at 4 am this morning (Sunday 15th) bound for Queenstown. It was listed as carrying four passengers. Weather conditions were classified as fair to good. Commercial flights do not commence until 6 am because of noise restrictions, and there was no other aircraft logged for arrival or departure at this time.

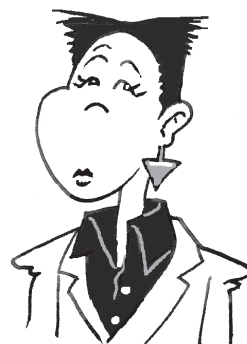
Dr George Donaldson University of the Northern City

Lightning could have produced the light and sound show that appeared in our northern sky on Sunday morning, but it didn't seem random enough to me. It could also have been an aeroplane. Personally, I think it was a more orchestrated aerobatic event. Only something very manoeuvrable could have produced the lights we saw.



A spokesperson for the Weather Bureau

Weather conditions were conducive to thunderstorm activity over the whole city during the early hours of Sunday morning, 15 July, and there may have been some incidents of lightning during this period. However, we have no confirming reports from our monitoring station in the northern suburbs.



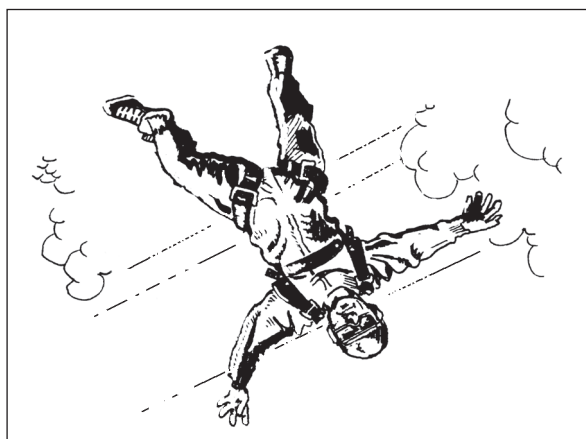
The Metropolis Times

Kingston Sky Spectacular

Hundreds of spectators will brave the anticipated cold and wet conditions tomorrow to see the beginning of the 25th Annual Australasian Skydivers Championships.

Spokesperson for the event, Mr Wally Gasgoine, said that this was going to be the most spectacular one yet, with almost 200 competitors likely to participate. Favourites for the 'All Comers' title is the four-person team of Grant Wilson, Ted Campbell, Wendy Johnston and Frans Vandenden, who will perform their world renowned 'sound and light' show in free fall from 5000 metres. 'We've been doing this together for years,' said Vandenden.

'It all began as a bit of a stunt. Grant's a real wag, and likes to upset the authorities, so we got together to leap off the Sydney Harbour Bridge carrying a banner that read 'Save the Yellow-Bellied Grass Shroot.' The environmentalists were right behind us until they discovered that there was no such creature. That's Grant for you.'



New Plane – Hush, Hush!

A leaked report has suggested that the government is in the process of purchasing a new stealth fighter-bomber to replace the ageing F-111. A government official denied that any decision had as yet been made about the Titan, however *The Metropolis Times*' copy of the report suggests that 'top secret' test flights of a prototype currently on a 'flying' trip to Australia, have already been carried out. It is believed that the Titan is currently in another capital city on the west coast of the country.

THE METROPOLIS TIME HOTLINE

Do you believe we have been visited by ETs?

YES: 131 2001

NO: 131 2002

Crosswords, Comics page 64

Letters to the Editor

The following letters are a small sample of the huge amount of correspondence we have received on this topic since the recent reports of a UFO over Metropolis last Sunday morning.

ET phone home

As far as we know, the nearest habitable planets are probably thousands of light years away. Why would ET come so far? He couldn't phone home even if he wanted to.

Mary Delacourt (Whopping Lane)

Why us?

Our family was driving home across the Simpson Desert when a UFO buzzed our car. The engine failed, and the car radio and car lights went out. We were really frightened. But why choose us? Why not the Prime Minister, or one of his scientists? Nobody believed us anyway.

Barry Smith (Wori Wori)

They're here!

Maybe now people will begin to believe that we have been visited by extra-terrestrials. These latest reports demonstrate that we are being watched by creatures far more intelligent than ourselves. Perhaps they're worried by our violent behaviour here on Earth. We should not be too complacent.

Molly Muggleweed (Grandville)

ET or not ET? – That is the question!

I'm tired of all this talk about extra-terrestrials. Any scientist will tell you that not only is there no proof of any such visitation, but the chances of ET stumbling upon our insignificant little planet stuck out in the backblocks of our Milky Way Galaxy is just too remote for words. Somebody ought to teach these thick-headed UFO devotees a lesson. They're so gullible.

Grant Wilson (Stanton)

Intelligent extra-terrestrials – I hope not!

History shows us that when technologically advanced races come into contact with primitive cultures, they invariably violently exploit them, no matter how noble their initial motives. Let's hope that we are not being visited and observed by intelligent extra-terrestrials. They may regard us as no more important than the ants.

Les Dennison (Grandville)

Northern lights – report

Write on the back of this page if you need more space.



Possible causes of the UFO sighting

Most likely cause of the UFO sighting

Reasons for your selection



In this scenario, the Chelmsurst Secondary College computer system has been struck down by a mystery computer virus that prevents any of the school's administration computers from doing anything else but display the number 19 on their monitors.

Despite the best efforts of the school's computer technician, he is unable to remove the virus without losing all the data on the school's server, including essential student and staff records. Fortunately, the hacker responsible for the virus has not yet unleashed its full destructive potential – preferring instead to set the principal and staff a series of puzzles. If these puzzles are successfully completed, the virus may be nullified, hopefully leaving the data intact.

The puzzles, which are mathematically based, also hold the clue to the hacker's identity.

The questions

- What are the solutions to the hacker's puzzles?
- Can the school's data be saved?
- Who is the hacker?

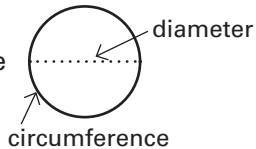
The support material includes:

- comments from the characters involved
- some background information
- a set of mathematical and logical puzzles to be solved.



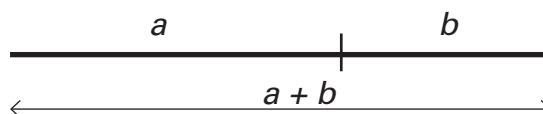
The mathematics of the mystery

Students will need to understand the significance of the following number concepts.

- **pi** (π) is the ratio between the circumference of a circle and its diameter, and is approximately 3.142. 
- **Prime numbers** are those that can only be divided by themselves and 1; for example, 2, 3, 5, 7, 11, 13, 17, 19.
- **Perfect numbers** are equal to the sum of their factors, excluding the number itself. For example, $6 = 6 \times 1 = 2 \times 3$. Discarding 6, its factors are 1, 2 and 3, which add up to 6.
- The golden ratio (ϕ , which is the Greek letter 'phi') is approximately 1.618, but is given accurately by the expression:

$$\phi = 1 + \frac{1}{\phi}$$

Described mathematically for a line, it is the ratio for dividing a line into two segments, a and b , so that the ratio of the entire line to the longer a segment is the same as the ratio of the a segment to the shorter b segment, as shown below.



The golden ratio has been used often in art and design since the 16th century, as it is considered aesthetically pleasing.

- The Fibonacci sequence is a mathematical series named for the Italian mathematician Leonardo of Pisa, who was known as Fibonacci. In this series, the terms are calculated by adding together the two previous terms: 1, 1, 2, 3, 5, 8 ... This series had been known before Fibonacci, but he is remembered for his mathematical explorations of the series.
- 'Googol' is the number equal to 10 to the 100th power (10^{100}), which means 1 followed by 100 zeros.
- Binary numbers – our numeral system (also known as the denary system, or base 10) uses 10 different numerals, 0 through to 9. The



binary system uses only two numerals, 0 and 1. The first 10 binary numbers are: 0, 1, 10, 11, 100, 101, 110, 111, 1000, 1001.

The binary system has been integral to computer design, as its two figures correspond to the two basic positions in electronic circuits – 'on' and 'off'.

- Two of the puzzles require an understanding of ciphers and codes.

DEBATE TOPIC

Computer hackers are no more than common criminals.

Background

This mystery focuses on one of the most serious problems for the computer age – the computer virus. These viruses exploit vulnerable computer systems by gaining unauthorised access, and often disrupting operations and corrupting files. Viruses are a source of great frustration and, in many cases, financial loss for businesses. Assisting businesses to develop adequate computer security systems has become a billion-dollar industry.

Unfortunately, with each improvement in security come newly evolved viruses; leading to a spiral of attack and defence, and spawning a new type of computer buff, the 'hacker'.

People committing burglaries are considered to be criminals, and the punishments can include imprisonment. The amounts stolen can usually be measured in the hundreds, perhaps thousands of dollars. In recent times, computer hackers inflicting viruses on the business community have cost companies many millions of dollars in lost revenue and general inconvenience.

Some hackers have been found to have had a malicious intent, but others have seen their actions as simply an adventure – to go where it is forbidden. Often, hackers are quite young, and this did result in an initially more lenient approach by the authorities. However, recently there have been calls for harsher punishments. This follows a similar hard line taken by those concerned about the proliferation of graffiti.

On the other hand, within the computer programming industry the term 'hacker' can be used to describe an extremely skilled and admired

computer programmer – not just a person who cracks a secure computer system.

Hackers also have their own code of ethics. Two of these ethics that have been in existence since the 1970s are:

- the belief that information should be freely available to all people, and so it is a hacker's duty to write free software and do what they can to facilitate access to information and computing resources
- the belief that system cracking for fun and exploration is ethically acceptable as long as the hacker commits no theft, vandalism or breach of confidentiality.

This debate can be extended to include so-called white-collar and victimless crimes.

RELATED ACTIVITIES

Drama

Students can act out the dialogue between the principal, vice-principal, the computer technician and the head of computer studies, and may also choose to write additional dialogue in which the hacker is interviewed by the principal.

The role of computers in society

This mystery can lead on to a more encompassing study of the changing role of computers in modern society. Businesses have become dependent on computers. Initially, the function of the typewriter was extended into what we call word-processing. Documents such as letters and account statements were prepared on computers, often as pro-formas. It was easier, and it saved time. Later the high-speed, data-processing capabilities of computers enabled companies to accumulate detailed records about their clients and their preferences, their company products and their sales performance. Business decisions became more informed, being based on more accurate and detailed information, stored and analysed on computers.

In the motor vehicle industry, for instance, computers have increasingly replaced people on the assembly lines and in the offices. But, has this dependence on the computer come at too great a cost? Are we more vulnerable? Will the terrorists of the future attack computer



systems rather than people? Are we becoming increasingly isolated from other people? For example, instead of talking to people over the telephone, we now increasingly interact with computers – providing opportunities for contact with people we have never met. Correspondence is increasingly undertaken via email, and business transactions through the Internet. People are even finding their partners over the net. Social interaction through chat rooms is increasing. Is this creating a society incapable of communicating face to face? Or is it widening people's horizons, allowing interaction with people from a variety of cultures and locations, with a corresponding variety of opinions and world views?. Where will this computer revolution finally take us?

Mathematics

A related study area is the subject of number. Ancient Greek mathematicians such as Archimedes, Pythagoras and Euclid set the foundation for much of modern mathematics, and make an interesting research topic. For example, the Pythagoreans (followers of Pythagoras) were regarded with suspicion by many in the population, and many were harassed and some even killed for their practices and beliefs.

Other mathematical research topics include:

- the history of pi (π), phi (ϕ) and the golden ratio
- the ancient link between mathematics and magic or mysticism, and even alchemy
- where the golden ratio is found in nature and in architecture
- why prime numbers are vital in Internet security
- the history of codes and ciphers, and their applications during wartime.

SAMPLE REPORT

Note: The marking scheme is a suggestion only. You may not want to score the work at all.

Record the responses required by the computer virus in the grid below. (6 points)

- | | | |
|----------|-----------|---------------|
| 1 prime | 2 square | 3 pi |
| 4 golden | 5 perfect | 6 one hundred |

What is the purpose of the seventh puzzle? What did it tell you? (5 points)

The equation $X_n = X_{n-1} + X_{n-2}$ defines the way in which Fibonacci numbers are derived (i.e. the sum of the two previous numbers). The first six Fibonacci numbers are 1, 1, 2, 3, 5, 8. Using these numbers by shading the first box on the first line of the grid, the first box on the second line, the second box on the third line, the fifth box on the fifth line and the eighth box on the sixth line, gives the name P Siler (one of the three suspects).

Explain how you arrived at each solution. (18 points)

- 1 Nineteen is a prime number, exactly divisible by itself and one.
- 2 The denary conversion of the binary number 0001101 is 25 – a square number (5^2).
- 3 The circumference (C) of a circle divided by its diameter (D) gives the constant pi (π) or 3.14159...
- 4 The equation $\phi = 1 + \frac{1}{\phi}$ gives the golden ratio, or phi (1.618...).
- 5 Six and 28 were considered by the ancient Greek mathematicians to be perfect numbers (they were the sum of their factors).
- 6 The six numbers are a simple substitution code (A=1, B=2, C=3, etc.) for the word 'googol', a term for the number 10^{100} .

Can the data be saved? How do you know?

My final comment is a bar graph code. Each column has a value between one and 26, and represents a letter of the alphabet. (1=A, 2=B, 3=C, etc.) Spaces between columns represent spaces between words. The code translates as:

The virus program has now deleted itself. Next time make sure you have a more secure firewall.

How would you punish the hacker if you were the school principal?

The punishment to be handed out to Peter Siler is a matter of individual judgement, but should be justified in the student's response to this question.



YOUR TASK – A computer virus has infected the Chelmsurst Secondary College's computer server. It manifests by showing the number 19 on the monitors of all the workstations connected to it. Neither restarting the computer nor running any of the virus detection and repair programs has had any effect. Even reloading the system files and running the backup restore program haven't helped. Erasing the server's hard drive is one solution, but with the backup files also infected, the loss of important student and teacher data makes this the last resort. Your job is to consider the comments of those investigating the virus, carefully examine the results of their efforts, and provide an explanation and solution to this problem.



This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

- 1 Read your task.
- 2 Read the extract of the discussion between the principal, the vice principal, the computer technician and the teacher in charge of computer studies.
- 3 Look up the meanings of any of the computer expressions in this mystery that are unfamiliar. Here is a list of some of these words:
 - hardware
 - software
 - system software
 - ROM
 - boot up
 - computer virus
 - hacker
 - trojan horse
 - firewall
- 4 Find out how to turn binary numbers (base 2) into denary numbers (base 10).
- 5 See if you can discover some mathematically famous and historically important numbers, of which 19 is an example.
- 6 Solve the hacker's puzzles and enter the answers (in words) in the grid on your report sheet.
- 7 Solve the seventh puzzle to find out the name of the computer hacker, and consider an appropriate punishment.
- 8 Solve the final code, the bar chart, to determine whether the school's files can be saved.



Fenella Lipman
School principal

Walter Harley
Vice-principal

Kelly Jackson
Head of computer studies

Jordan Pagonas
Computer technician

Extracts from a discussion that took place between Fenella Lipman the school principal, Walter Harley the vice-principal, Jordan Pagonas the school's computer technician and Kelly Jackson, subject teacher and head of computer studies.

FENELLA Well Jordan, what can you tell us about our computer problem?

JORDAN Not a lot at this stage Mrs Lipman, other than it's obviously a computer virus, and it has both hardware and software components.

FENELLA What does that mean?

JORDAN Well, reloading the server's system software makes no difference. On boot-up, control is taken over by one of the system ROMs, which prevents a normal startup and places the number 19 on the screen of any workstation connected to it.

WALTER Why not simply replace the server motherboard?

JORDAN I thought of that, but my virus detection software has discovered signs of a Trojan horse.

WALTER So you're worried that if you change the motherboard, the Trojan horse will be activated on boot-up and corrupt everything.

JORDAN That's right. I think that the renegade ROM is designed to keep the Trojan horse in check.

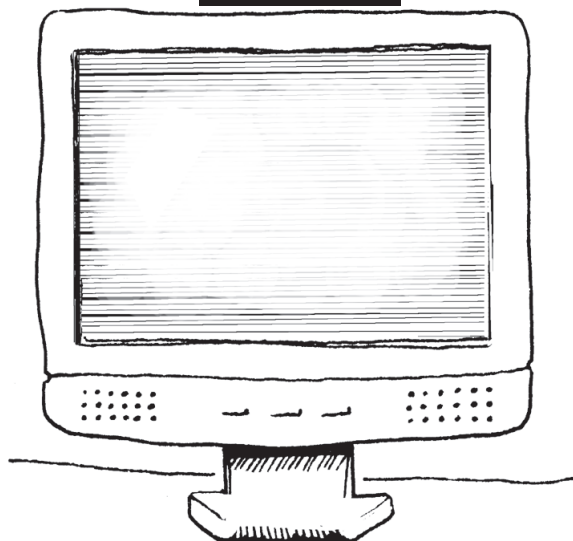
FENELLA So what you're saying is that we're going to lose our files no matter what we do?

KELLY Not necessarily. I've been wondering why the virus didn't simply destroy all of our files. Apparently it hasn't, since Jordan tells me that the data files appear to be intact. So I suspect that our hacker is playing games with us. Perhaps he, or she, doesn't necessarily want the school files destroyed, just wants to make us jump through hoops. I think our hacker has a low opinion of our computer security and our problem-solving ability, and has set a puzzle for us to solve. I think, if we are smart, there is a way out of this.

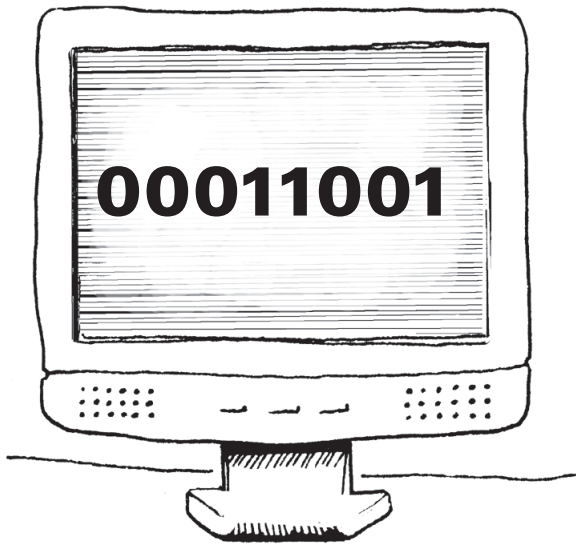
- FENELLA** You mean that if we play along with this infantile and irresponsible game, our system may survive unscathed?
- KELLY** Perhaps. But you do need to accept the possibility that even if we are successful, our hacker may still wipe the files. But I can't see that we have any other option at this stage.
- WALTER** What about suspects? Are there any students with the computer knowledge to pull this off?
- KELLY** Only three that I can think of – Aroha Te Puea, Van Nguyen and Peter Siler. They hang around together, and I guess you'd have to say that they fit the stereotype of 'computer nerds'. They're always on their computers, either programming, gaming or surfing the net.
- WALTER** Have you questioned them about the virus?
- KELLY** Yes, it was the first thing I did. But when I suggested that no one else at the school has the skills to cause this much trouble, they all assumed that hurt look of the unjustly accused! Unfortunately, I don't think we're going to get anything out of them. I strongly suspect that this is the work of only one person. I don't think that it's a collaborative effort, and I don't think that the other two are aware of the identity of the guilty party – assuming it is one of these three.
- FENELLA** Well, I think we have to assume that at this stage. So how do we begin?
- KELLY** I think that we work out how to respond to the number 19.
- FENELLA** I see. You're saying that our hacker expects a keyboard response to 19, and if we get it right there may be another puzzle, and another, until he or she thinks we've done enough and goes away?
- KELLY** Yes, that's what I suspect will happen, and I think we may have to do some research to be able to complete these puzzles.
- FENELLA** Okay, start by telling me what you know about 19.

From this point on, the four of them match their wits against the hacker, looking for each correct response as the virus program produces one puzzle after another. Their aim is to successfully regain control of the server and save their files, but also, if possible, discover the hacker's identity. Kelly Jackson knows a little about the personality profile of hackers – often they are quite arrogant and confident of their superior abilities. This can result in their consciously or subconsciously leaving clues to their own identity somewhere within the coding, just as a graffiti artist will sign their work.

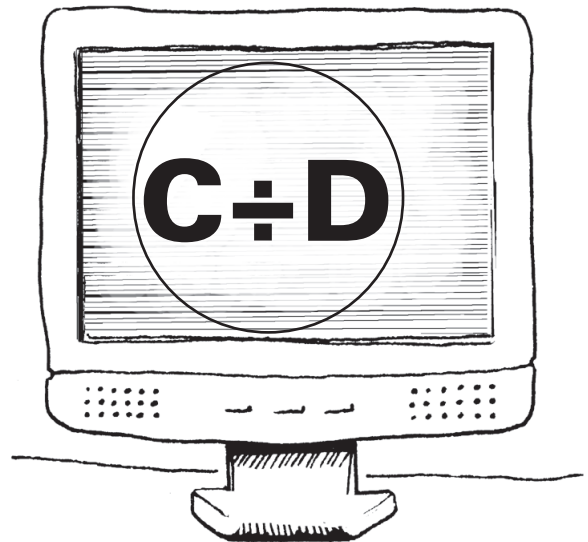
PUZZLE 1



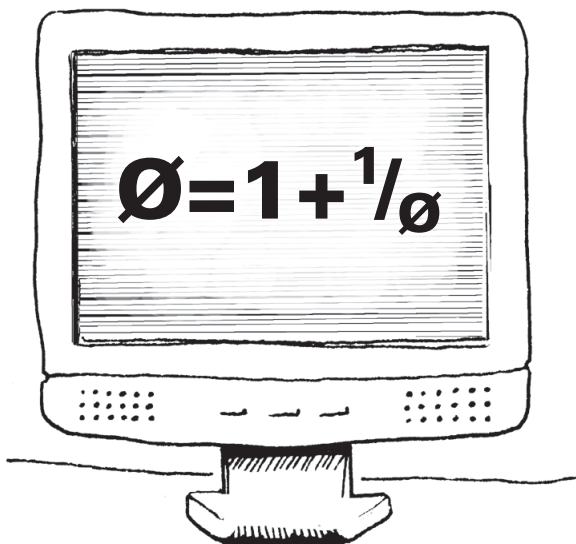
PUZZLE 2



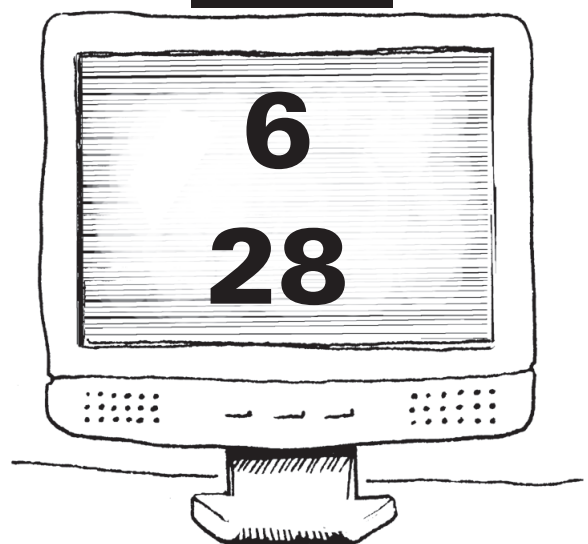
PUZZLE 3



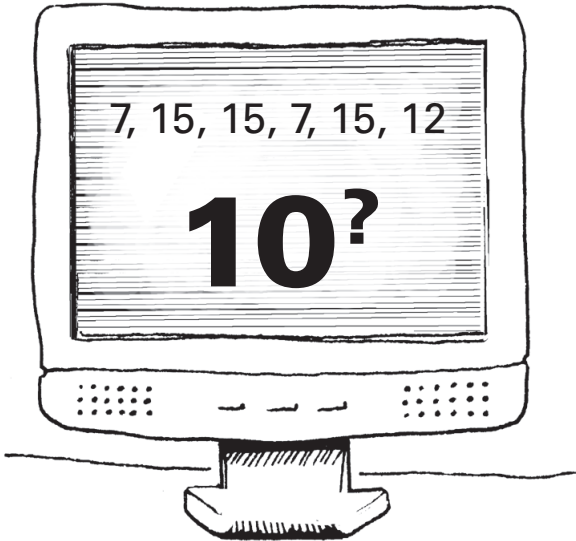
PUZZLE 4



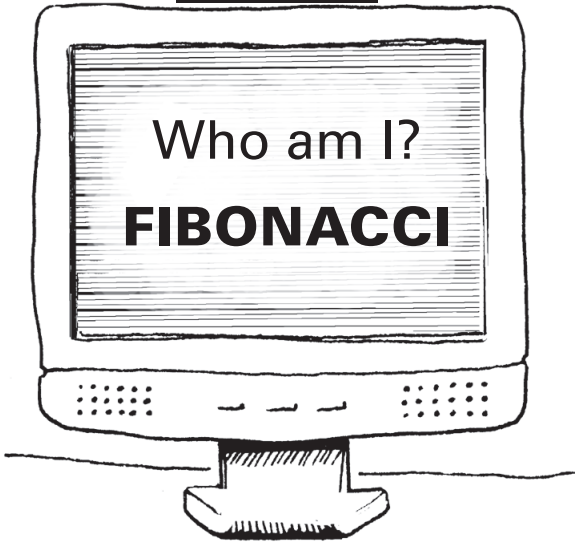
PUZZLE 5



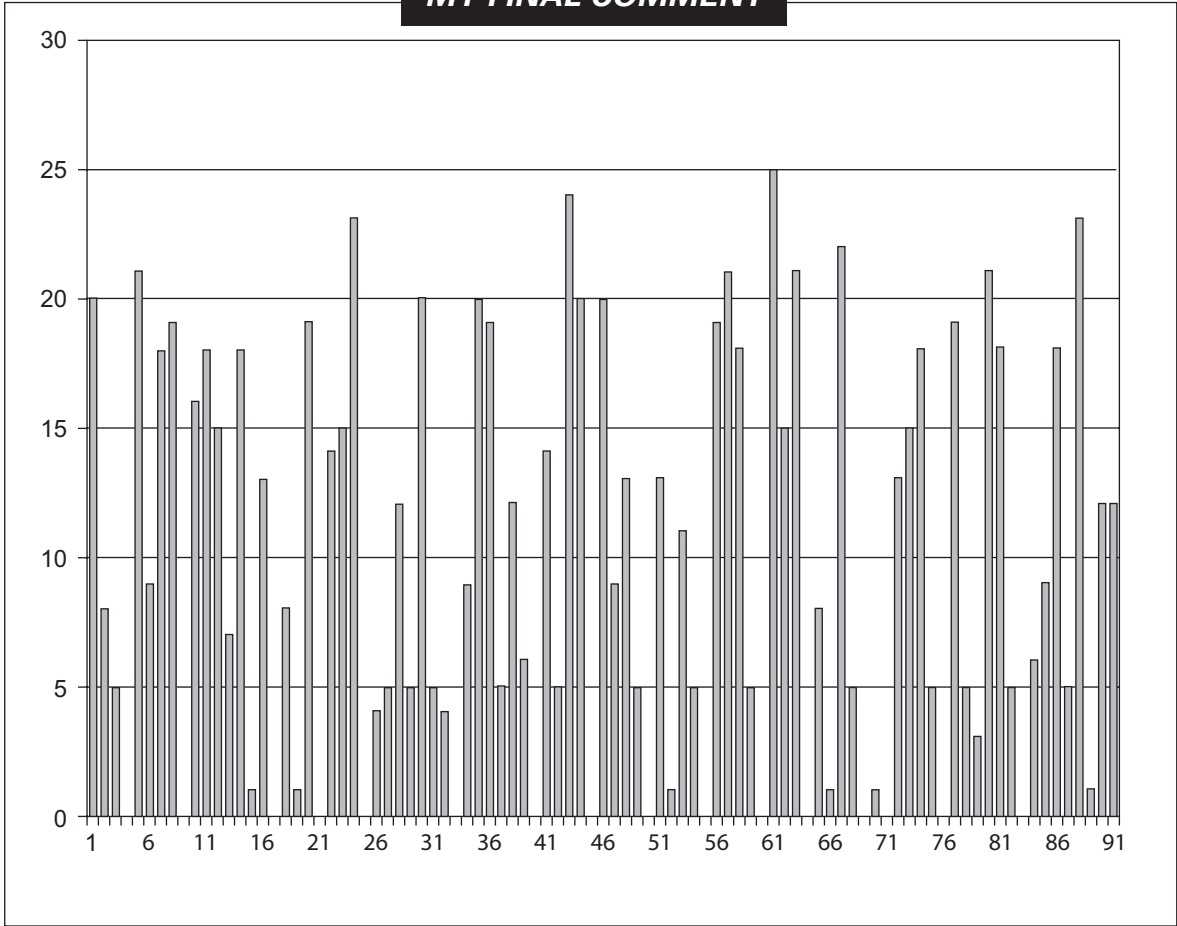
PUZZLE 6



PUZZLE 7



MY FINAL COMMENT



Nineteen - report

Write on the back of this page if you need more space.



Record the responses required by the computer virus in the grid below.

1									
2									
3									
4									
5									
6									

What is the purpose of the seventh puzzle? What did it tell you?

Explain how you arrived at each solution.

1

2

3

4

5

6

7

Can the data be saved? How do you know?

How would you punish the hacker if you were the school principal?

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